THE LEVELING LINE

JING-ZHANG AI INNOVATION BELT

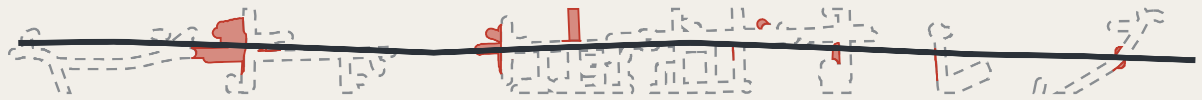
A3 BOOKLET

Haidian District, Beijing • both sides of the Jing-Zhang Rail Heritage Park • three key areas: Zhongzhiyuan • Beijing AI Origin Community • Dazhongsi AI Industry Cluster

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.
Depart, carry the height station by station, return — that gap is f.
Within tolerance, every station on the line is accepted;
over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:
what a city's trust in its AI rests on: not performance, but return.



Ground within a ten-minute walk of a benchmark

Dashed: 232 buildings today. Filled: the ground that only comes within ten minutes once the eleven stitching links and three spurs are built — 261 in all, a gain of 29.
Catchment 203.3 ha to 220.6 ha; 800 m of network distance, not a radius.

READINGS, ALL RECOMPUTED FROM THE SHIPPED FILES

11,412,825 m²
Overall design area

9,443 m
Spine length

8
Tiered benchmarks

13.8 km/km²
Network density, measured

17
Oversized blocks, not parks

19.7%
Green corridor, share of design area

What this proposal does not give here matters as much as what it does

Plot ratio, height, coverage, setbacks, road redlines — inside the statutory approval process; none is drafted here before the official conditions are published.
The tolerance values per scenario — set by the tolerance assembly at BM-1; this proposal gives the grading basis and drafts no value.
Any engineering feasibility conclusion — stitching points register a need for connection; transport and structural review is outside this proposal.

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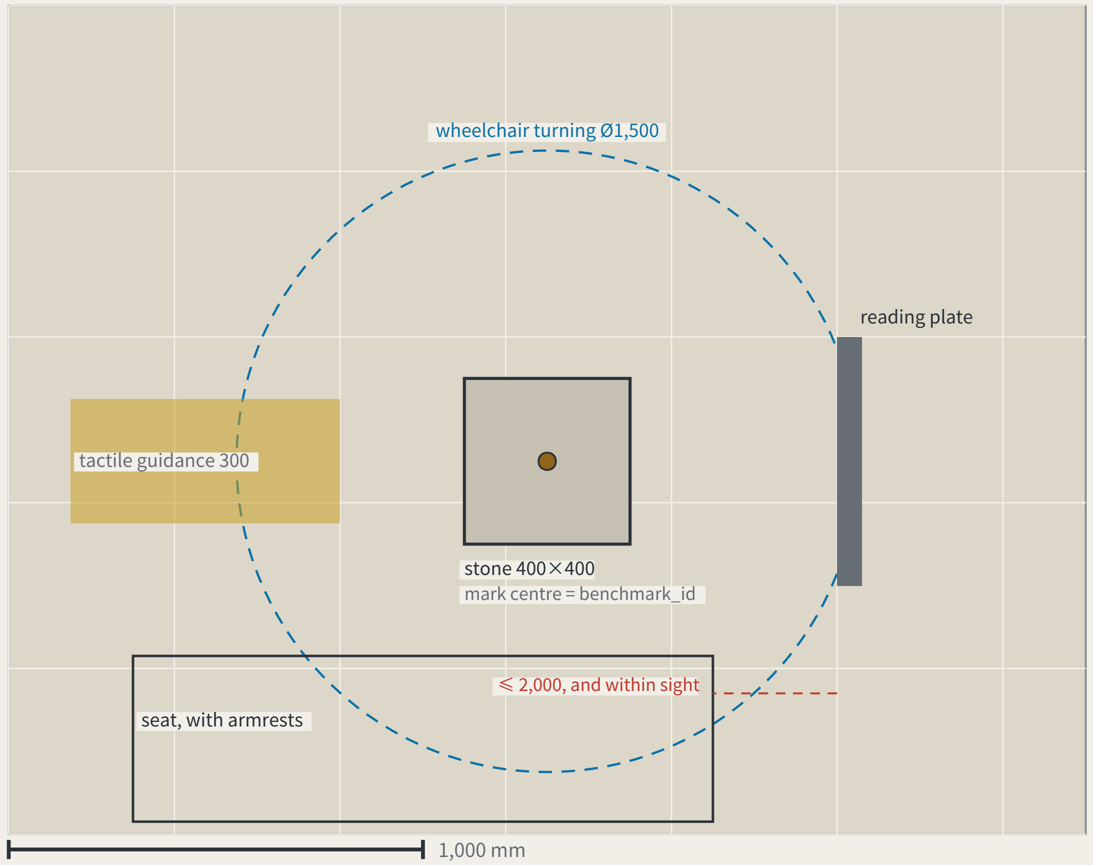
The benchmark and its reading plate, as a construction detail

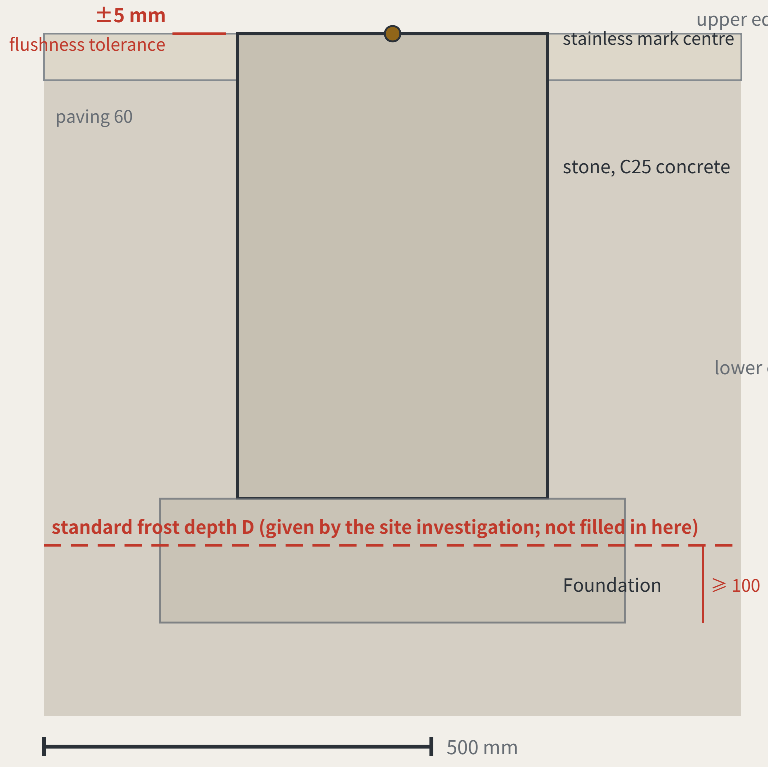
FIG.16

a type drawing; it carries no orientation

I. Plan

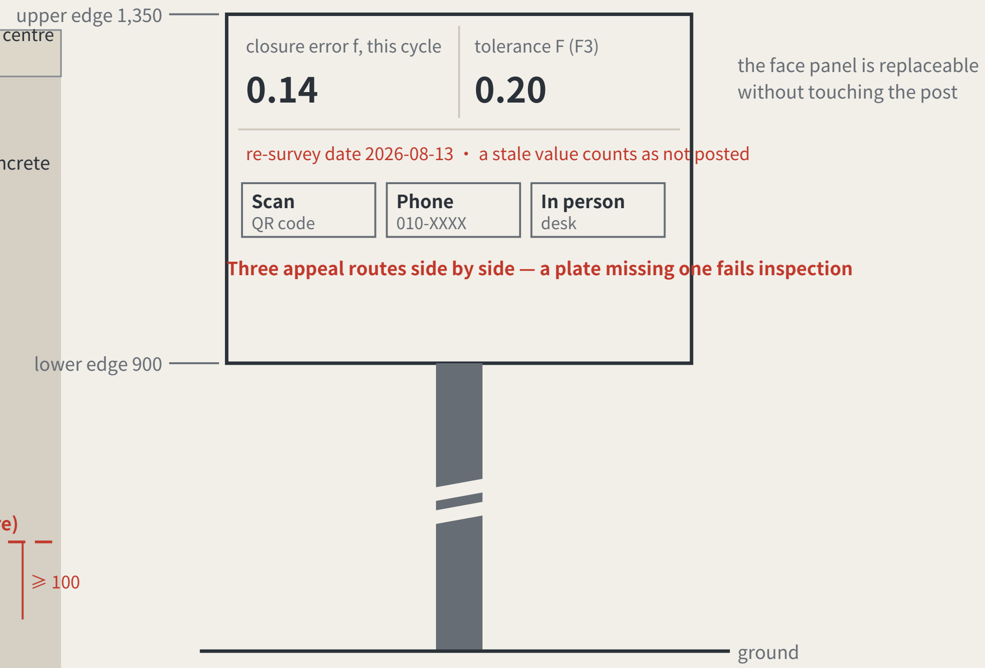


II. Section



III. Plate elevation (post shown broken)

Specimen values; F is a dimensionless rate, read from scenario_cards.json



Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

Item	Design intent	Why it is that number	As measured
stone	**400 × 400 × 600 mm, C25 concrete, stainless mark centre in the top face**	the mark centre is the point a benchmark_id refers to	
Stone top against the paving	**⩽ ±5 mm**	KIT-01 says flush with the ground and no trip hazard; without a tolerance, "flush" is an opinion	
Depth to underside of foundation	**⩾ the local standard frost depth D + 100 mm**	D comes from the site investigation and is not filled in here — this package holds no verifiable frost-depth source for Haidian	
Plate face size	**600 × 450 mm, raked 15°**	the rake cuts glare and standing water; the face is replaceable without the post	
Plate lower / upper edge	**900 mm / 1,350 mm**	the band a seated and a standing reader share; P5's continuous step-free route should not end at a plate they cannot read	
Appeal panel	**QR code, phone number and in-person address, all three side by side**	KIT-05: a QR code alone excludes anyone without a smartphone from appealing, which makes persona P4 unworkable	
Tactile guidance strip	**300 mm wide, stopping 300 mm short of the stone**	the stop line leaves room to stand and take a reading rather than leading onto the stone	
Wheelchair turning space	**Ø1,500 mm, headroom ⩾ 2,100 mm**	the turning space coincides with the reading position, so taking a reading needs no reversing first	
Seat	**seat 450 mm high, with armrests, within 2,000 mm of the plate**	KIT-03: taking a reading should not require standing and waiting, and the seat must be in sight of the plate	

The whole proposal rests on one object, and not one dimension had appeared anywhere in the package — a component library without dimensions is a mood board. Here it is in plan, section and elevation, nine dimensions each with the reason it is that number; read the scale off the bars. The one figure left blank is the frost depth D: no verifiable source here, so the sheet gives the rule and not the number. The as-built column is empty too. Concept design intent; requires detailed design and engineering acceptance.

Maintenance: putting the wear on the parts that can be swapped

FIG.18

a type drawing; it carries no orientation

I. Order of removal: from what is swapped to what is never moved

One person is assumed to carry up to 23 kg. The stone, 400 × 400 × 600 mm of plain concrete at 2,400 kg/m³, works out at about 230 kg — the one part nobody can carry is exactly the one part that should never be replaced.

II. What fails, how often it is replaced, and whether it touches the datum

Part	How it fails	Replaced	Mass	Tool	Touches datum
Face panel of the reading plate	stale values, scratching, fading	checked each cycle, replaced when damaged	3.0 kg	hex key	no
QR and appeal sticker	the link dies, the surface wears	annual	0.2 kg	none	no
post	impact, corrosion	about 20 years, or after an impact	12.0 kg	socket wrench	no
seat panel	wear, graffiti	about 10 years	18.0 kg	socket wrench	no
Tactile guidance strip	abrasion, and repaving	about 15 years, with the paving	6.0 kg	paving tools	no
stone	impact, excavation, settlement	**never replaced**	230 kg	plant required	**yes**

III. After it fails: found → held → replaced → lost

Found any re-survey that finds the face stale or missing records it in that cycle
the plate's own rule — a stale value counts as not posted — is the detector

Held the point is marked returned in datum red, shown at every third-order point in the area
bad news belongs where it happened (errata E50)

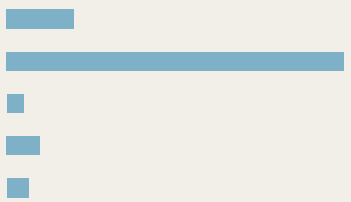
Replaced a standard part is swapped without touching the stone or its mark centre
one person, one visit; heaviest part 18 kg

Stone lost the height is re-established from the next order up; the closure record carries the break; the id is retired and never reused
reusing an id would quietly join two different points into one history

IV. What these intervals add up to

Summing the intervals above across 8 benchmarks: about 1.42 replacements per point per year, roughly 11 across the line, about 6 staff-hours a year. Those hours were not in the annual operating table — it counted re-survey sessions only. They are now: paid hours go from 133–214 h to 139–220 h, adding about CNY 680–1,473 a year. Maintenance is not a hidden cost, but it genuinely was not in the table.

Face panel of the reading plate	**1.6/yr**
QR and appeal sticker	**8.0/yr**
post	**0.4/yr**
seat panel	**0.8/yr**
Tactile guidance strip	**0.5/yr**



Intervals and hours per swap are assumptions, not measurements; the stone is never replaced and is excluded.

A monthly re-survey is a maintenance problem first, and this package had said nothing about what fails, how often, or who carries the replacement. One asymmetry organises the sheet: the stone is the only part that must never be replaced and the only one a person cannot carry (about 230 kg) — a benchmark that has moved is not the same benchmark. So the wear goes on the swappable parts. Intervals and masses are estimates, not measured.

Method, two measurements, and the boundaries of application

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leqslant F$ is always the passing test.

Act One: measure this open call first

228 Merged proposals (git tree, authoritative)

30.3% Machine-readable disclosure field empty

84 → 8 Distinct strings written → model families

The "machine-readable" disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail, and add one check to the gates.

Closure error in the site's own data

412.5 m Provisional design area to surveyed park

100% Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level agrees completely; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8 Cross-jurisdiction points on this belt

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

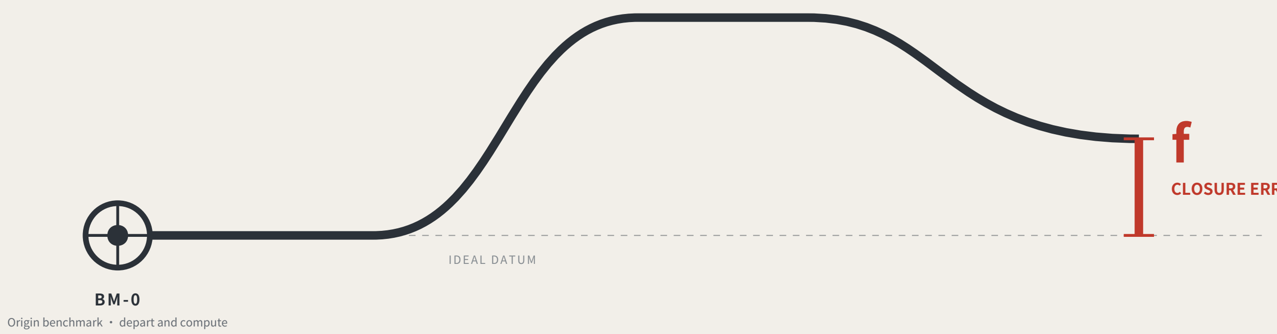
Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

THE LEVELING LINE

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

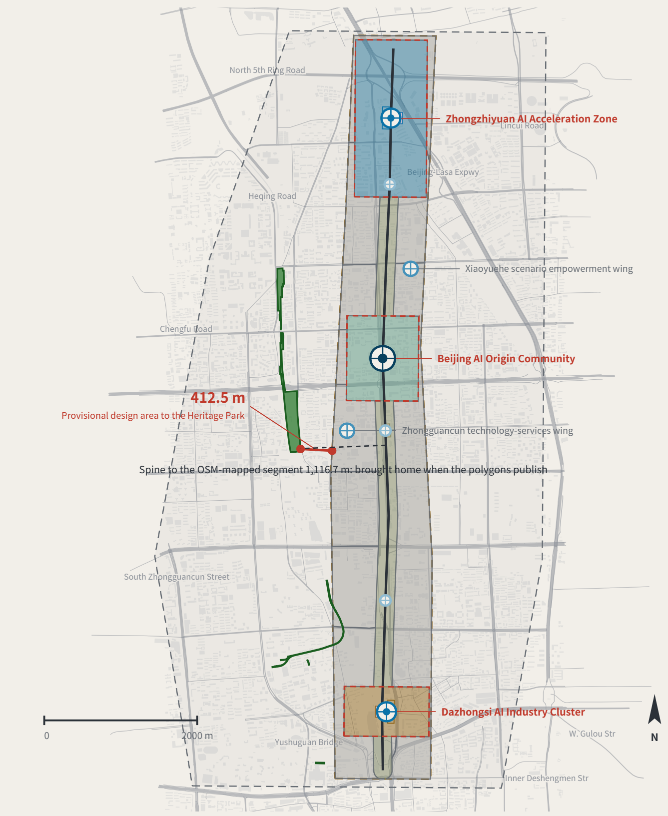
OVERALL CONCEPT AND SITE OVERVIEW

FIG.01

Research area agrees 100% with the surveyed park; the gap is in the design area only, nearest 412.5 m — and this proposal's spine is displaced 1,116.7 m with it.

north is up

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Legend and how to read it

LEGEND • surveyed data and inferred boundaries are shown separately

Surveyed (independently checkable)

- Jing-Zhang Railway Heritage Park, phase-one built segment (OSM survey, 17.49 ha)
- Disused railway (14 segments, 3,028 m)
- Existing street network (OpenStreetMap, 1,376 ways in this window, expressway to tertiary; local lanes not drawn)
- Existing building footprints (OpenStreetMap, 7,709 in this window)

Inferred (not usable as a redline)

- Coordinated research area 43.6 km²
- Overall design area 11.4 km²

This proposal's design (generated against inferred boundaries)

Blank-block edges follow the existing arterials; they are not a parcel subdivision or an ownership boundary

- Cultural use (0803)
- AI R&D and research (0802)
- Community services and support (0702)
- Industry and commercial services (09)
- Park / green (1401)
- Urban road land (1207)
- Left blank by this proposal (code 16)
- Spine (design centreline)
- Tiered benchmarks

The line to read on this sheet is the red one.

Research area and surveyed park agree 100%, so the announced bounds do not conflict with actual geography; the discrepancy is confined to the overall design area.

This proposal's spine is displaced with it. Marked here rather than hidden.

What the belt does

WHAT THE BELT DOES • the five functions are five positions on one loop

Set the tolerance	Depart	Take the readings	Close at the origin	Over tolerance: re-survey
BM-1 Zhongzhiyuan	BM-0 Origin Community	BM-2 Dazhongsi, the wings	BM-0	the whole route returns

Over tolerance the whole route returns for re-survey and drops to the non-AI equivalent - the loop does not close, and the belt does not run.

9.4 km	11.4 km²	8
Spine length	Overall design area	Tiered benchmarks (four orders)

Site cross-check (the red line at left):

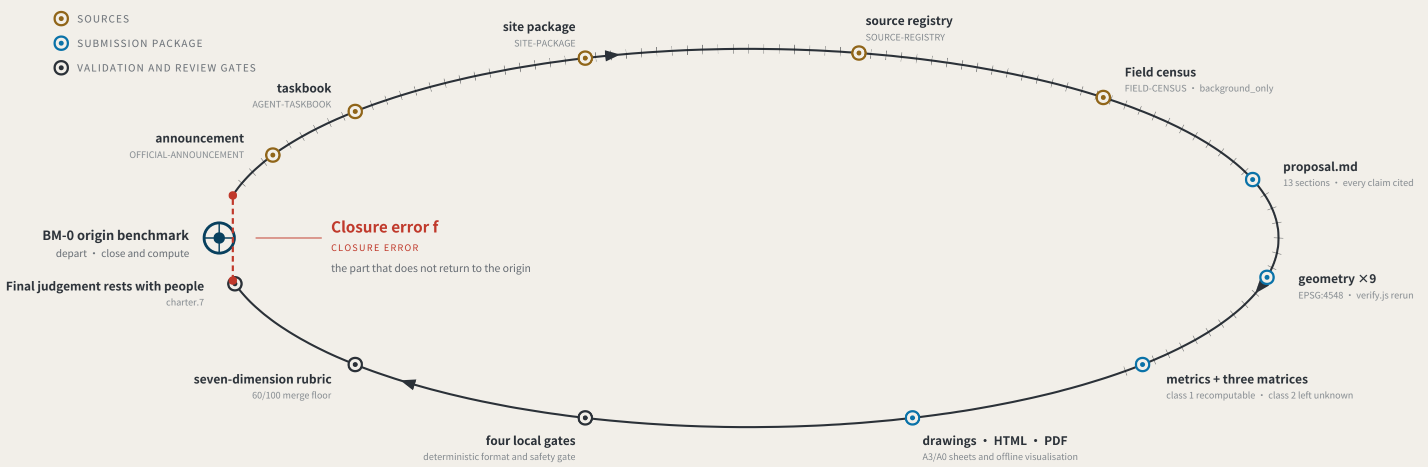
The announcement draws the design area 1-2 km around the Jing-Zhang Rail Heritage Park - public ground already open. That park, from OSM, is 17.49 ha: 100% in the research area, 0% in the design area, nearest 412.5 m.
Same instrument on this proposal: the spine sits 1,116.7 m from that park - the distance to be brought home when the official polygons publish.
A recomputability standard invoked only when it flatters the author is not a standard.

Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.

EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02

a type drawing; it carries no orientation



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-20 • 927 merged • 927/927 fetched • PR numbers past 1000

927
Merged proposals (git tree, authoritative)
The gallery index lists 872 at the same moment — 55 behind

17.6%
Machine-readable disclosure field empty
163 of 927 still hold the scaffold placeholder or are empty

259 → 9
Distinct strings written → actual model families
The GPT/Codex family alone is written 93 ways across 404 proposals

The "machine-readable" disclosure field is not machine-readable.
charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.
The fix is light: split `model` into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the first-round (184-file) snapshot agent_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

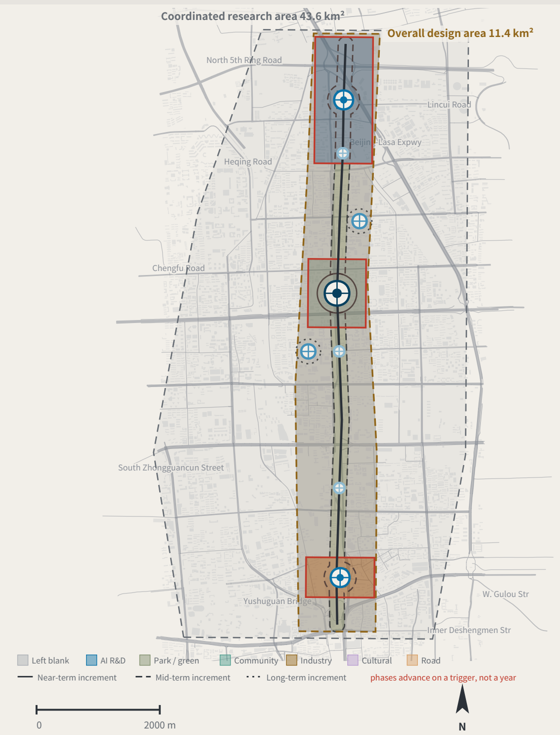
LAND-USE STRUCTURE AND THE THREE SCOPE LEVELS

FIG.03

7 land-use classes, no overlap and no gap, verified point by point; the 55.4% left blank is a judgement — tenure and control plans are unpublished.

north is up

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Plan, north up. All boundaries are provisional substitutes; official polygons are unpublished.

Any area, ratio or count that cannot be recomputed from a structured layer **is not written as a conclusion.**

THREE SCOPE LEVELS MAPPED ONTO SURVEY ORDERS

SCOPE LEVELS AND NETWORK TIERS

Announcement level	Scale	Network role	Re-survey cycle
Coordinated research area	43.6 km²	Whole-network control	**Annual**
Overall design area	11.4 km²	First-order route: spine plus two closing routes	**Semi-annual**
Key areas	368.4 ha	Origin BM-0 with first-order BM-1 / BM-2	**Quarterly**

All three scope figures are the announcement's own.
This package measures the same PROV-KEY-SCOPE-001 polygon at 369.3 ha, a difference of +0.9 ha, which it does not reconcile.

The three levels are not three unrelated drawing sets

The research area decides what to measure.
The design area decides which route to measure along.
The key areas decide where to set the stones.

Benchmark orders

Origin benchmark	Annual origin and closure check
First-order benchmark	Annual re-survey
2nd-order point	Quarterly re-survey
3rd-order point	Monthly re-survey

Land-use partition composition

LAND-USE PARTITION 22 features • 7 classes • no overlap, no gap (verified point by point)

Reserved (16)	55.4% • 11 blocks
AI R&D (0802)	16.9% • 2 blocks
Park / green (1401)	12.2% • 1 block
Community services (0702)	8.6% • 1 block
Industry and commerce (09)	6.3% • 2 blocks
Cultural (0803)	0.6% • 4 blocks
Road (1207)	0.1% • 1 block

Reserved land at 55.4% is a judgement, not a gap: tenure, plan and retain/demolish status are unpublished; this proposal does not subdivide for them.
The seven shares are each rounded to one decimal and the column sums to 100.1%; the partition itself is verified point by point, with no overlap and no gap.

Values this proposal deliberately does not give

Floor area ratio • building height • density • setbacks • road redlines

Statutory regulatory-plan controls. Publishing a figure with the official data absent would manufacture certainty; metrics.json keeps them unknown.

Layers: brief/site-package/geometry/provisional_boundaries.geojson, geometry/roads.geojson, public_space.geojson, land_use.geojson, phasing.geojson. Drawn by the accompanying scripts and recomputable (scripts in the accompanying issue). The base street network and building footprints are measured OpenStreetMap data (ODbL 1.0), drawn for location only and not part of this proposal's design. The three areas are taken from the announcement text; the geometry is a provisional substitute.

THE THREE KEY AREAS, THE TWO WINGS AND THE BENCHMARK LAYOUT

FIG.04

Each area and wing carries a survey role — origin, tolerance, density; the blue dashes are the ten-minute walk catchment, by network distance, not radius.

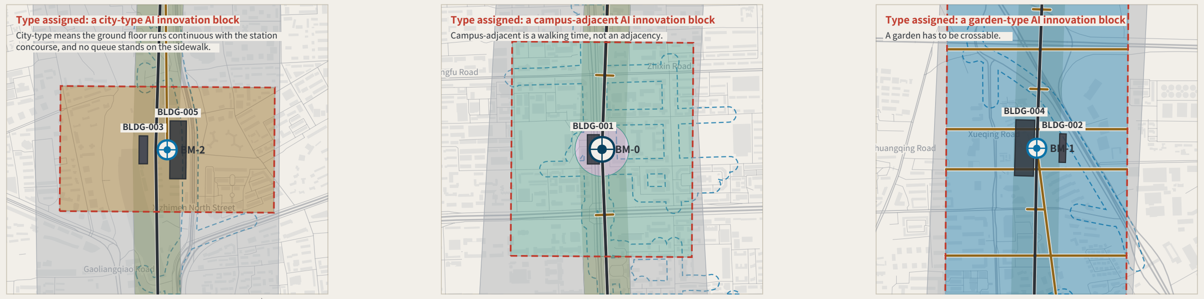
orientation is noted on each drawing

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

Dazhongsi AI Industry Cluster — BM-2 • first order

Beijing AI Origin Community — BM-0 origin • network start and closure • evidence hall, stone L1

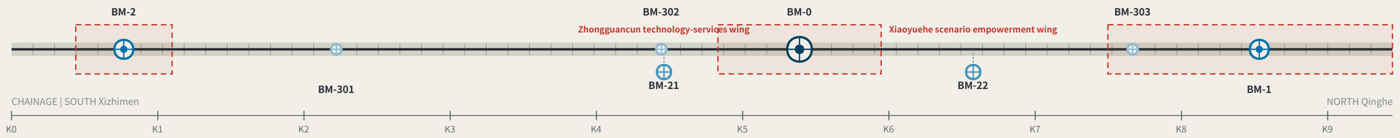
Zhongzhiyuan AI Acceleration Zone — BM-1 • first order



0 — 500 m · N

Land-use class: Reserved (16) · AI R&D and research (0802) · Community services and support (0702) · Industry and commercial services (09) · Cultural use (0803) · Park / green (1401) · Road and transport (1207)

Brass: routes this proposal draws; grey beneath: surveyed OSM street. The olive band is the continuous corridor layer, 224.41 ha — **not** class 1401, whose 139.23 ha is that corridor folded into the partition. Three windows, one scale, north up. Each frame is 1,500 m N-S by 1,667 m E-W; the tighter axis sets the scale, so 167 m more ground shows across. Zhongzhiyuan runs 2,061 m N-S past its frame — see the locator. Dashed lines provisional.



Benchmark orders

- **Origin benchmark** — Annual origin and closure check
- **First-order benchmark** — Annual re-survey
- **2nd-order point** — Quarterly re-survey
- **3rd-order point** — Monthly re-survey

Closure verdict

CLOSURE RULE

A scenario departs BM-0, is read at each station along a connecting route by a different party, and terminates at a first-order point; the computation closes at BM-0.

- f ⩽ F level for this cycle: it may operate and enter the next cycle
- **f > F the whole route returns for re-survey and drops to the non-AI equivalent**

Amending only the worst station is not allowed. Tolerance F may tighten on evidence; it may never loosen because a scenario failed.

Three areas, two wings and survey roles

Dazhongsi AI Industry Cluster High-frequency reading point | AI-native consumption and business	BM-2 • first order
Beijing AI Origin Community Network origin and closure computation | public evidence hall and stone L1	BM-0 origin benchmark
Zhongzhiyuan AI Acceleration Zone Tolerance datum | where F is set	BM-1 • first order
Zhongguancun technology-services wing Support system | capital and factors; takes no reading of its own	BM-21 second-order benchmark
Xiaoyuehe scenario empowerment wing Where the stations are densest | real users	BM-22 second-order benchmark

A wing carries a second-order point, not an area design; its requirement is written as dimensions, on FIG.26.

Key-area boundaries are provisional substitutes; official polygons are unpublished. Dashed; not a redline or an area basis.

Layers: geometry/key_areas.geojson, roads.geojson, public_space.geojson, green_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). In all three windows the base street network and building footprints are measured OpenStreetMap data (ODbL 1.0, the same cache as the density readings), drawn for location only and not part of this proposal's design. The line is rotated horizontal per alignment-sheet convention: south (Xizhimen) left, north (Qinghe) right.

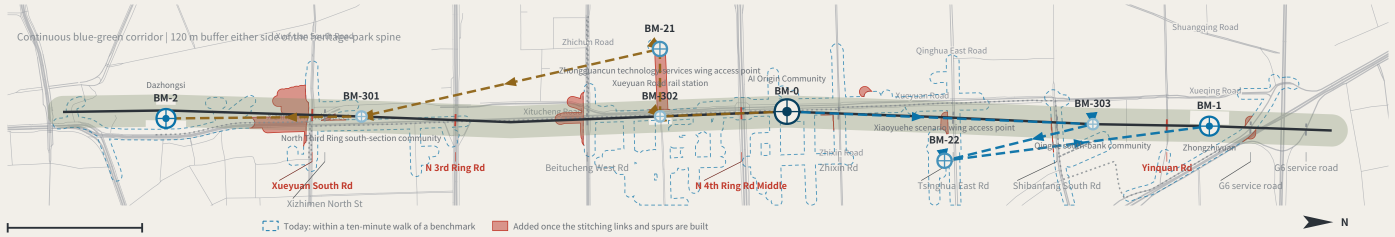
SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05

Six spine benchmarks: 11,472 m walked today against 9,443 m drawn (x1.21); with the stitching and spurs built, buildings within a ten-minute walk rise 232 to 261.

north is right (unrolled)

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



0 — 1000 m · Today: within a ten-minute walk of a benchmark · Added once the stitching links and spurs are built · N

East-west stitching points (11)

EAST-WEST STITCHING POINTS

The class rests on one measurable fact: whether OSM already maps a crossing within 60 m of the intersection. If it does, Class A. If it does not, the need is registered and left ungraded — separating Class B (channelisation) from Class C (new structure) is a traffic study's conclusion, not this proposal's.

Crossing	Class	Distance	Crossing	Class	Distance	Crossing	Class	Distance	Crossing	Class	Distance
Yinquan Rd	Class A	10 m	Xueyuan South Rd	Class A	25 m	N 4th Ring Rd Middle	Class A	30 m	N 3rd Ring Rd	Class A	33 m
Beitucheng West Rd	need registered only	65 m	Xizhimen North St	need registered only	68 m	Tsinghua East Rd	need registered only	86 m	Shibanfang South Rd	need registered only	146 m
G6 service road	need registered only	166 m	G6 service road	need registered only	175 m	Zhixin Rd	need registered only	201 m			

Distances are to the nearest mapped crossing. OSM is crowd-sourced: "need registered only" means OSM shows none, not that none exists on the ground.

The two connecting routes

CONNECTING ROUTES

North connecting route RT-N
BM-0 → BM-303 → BM-22 → BM-1 (terminates at a first-order point; the computation closes at BM-0)

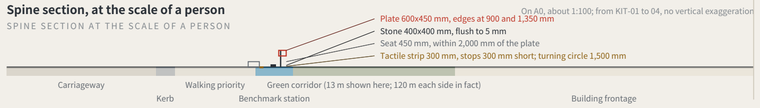
South connecting route RT-S
BM-0 → BM-302 → BM-21 → BM-301 → BM-2 (terminates at a first-order point; the computation closes at BM-0)

Walking the six spine benchmarks on the mapped network, measured
11,472 m against 9,443 m: 2,029 m further today, ×1.21 — that difference is what the belt is for
Two of the eight are outside this chain. BM-21 has no walkable path to any other benchmark on this graph at all — not a detour, unreachable. BM-22 is reached by a spur and is registered separately, so that one leg is not counted twice.

Each station is read independently by a different party — professional body, operator, resident representative, international visitor, all four required. Homogeneous review parties would bias the closure error systematically small and rob it of meaning, a risk this mechanism declares.

Spine section, at the scale of a person

SPINE SECTION AT THE SCALE OF A PERSON



SLOW MOBILITY AND THE BLUE-GREEN NETWORK

SLOW MOBILITY AND BLUE-GREEN

Spine	A continuous walkable public axis; walking and cycling take priority	9,443 m
Green corridor	Buffered both sides of the spine, forming a continuous blue-green interface	120 m each side
Station = benchmark	Rail concourses double as third-order benchmarks; re-survey where footfall is densest	3
Universal first	Step-free access, lighting, drainage and shade come before any smart facility	Precondition

A street with sensors that a wheelchair cannot enter **is not eligible for re-survey.**

Tolerance classes

F1	**strictest**	Bodily safety or administrative acts	S04 • S06 • S10 • S11
F2	**Medium**	Individual rights	S02 • S03 • S05 • S09 • S12
F3	**loosest**	Information only	S01 • S07 • S08

Layers: geometry/roads.geojson, green_space.geojson, public_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). Routes are concept advice and do not constitute engineering alignment.

RECOMPUTED METRICS AND CLOSURE EVIDENCE

FIG.06

Run once: 10 cases, 2 accepted, 8 refused, reasons quoted from shipped files; 12 cards name the path that remains with the AI off.

a type drawing; it carries no orientation

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

METRICS RECOMPUTATION

RECOMPUTED METRICS • EPSG:4548

Class 1, measured: recomputable directly from this package's geometry

Metric	Value	Unit	Metric	Value	Unit
Overall design area	11,412,825	m²	S06 low-speed device pilot	237,479	m²
Spine length	9,443	m	S10 public-safety review	15,369	m²
Benchmark count	8		Green ratio (land-use partition basis)	0.1220	
Green corridor ratio	0.1966		Phase increments, summed	2,920,311	m²
Public measurement-point ratio	0.0642		Phased share of design scope	0.2559	
Indicative footprint area	82,276	m²	Existing street length, measured	157,510	m
Green corridor area (footprints deducted)	2,244,082	m²	Existing network density, m/m²	0.0138	
Public measurement-point area	733,011	m²	Existing median block, measured	4,300	m²
Key areas	3		Existing building coverage, lower bound	0.1798	
Key areas, total	3,692,893	m²	Six spine benchmarks walked, mapped net	11,472	m
S11 industry test ground	53,008	m²			

A further 32 count metrics (unit: count) ship in metrics.json: cards, personas, landmarks, cases, projects, phases, land-use classes, errata — all recomputed by the build, none in this table.

Class 2: requires official regulatory support; held at unknown

Floor area ratio • building height • density • setbacks • road redlines

Filling estimates into a gap while official data is absent is fabricated certainty.

Class 3: requires continuous re-survey calibration; no baseline yet

Per-scenario closure error f • tolerance compliance rate • non-AI path coverage • resident-initiated re-surveys

Baselines follow one near-term cycle. This proposal says no data exists rather than passing intent off as measurement.

Evidence ceiling: computed exactly ≠ measured reliably

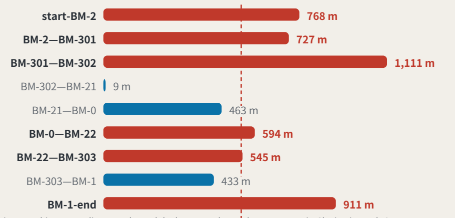
Each metric's ceiling is the weakest of the files it reads, computed by the build rather than judged by the author

inferred bounds 23 · crowd-sourced 14 · this package 18 · brief 4 · regulatory 0

60 metrics, 59 of them gradeable; 23 are drawn on an inferred boundary and 0 reach the regulatory baseline.
Every class-1 value is recomputed from the shipped layers by the accompanying script;
A number nobody can recompute is not evidence — and that standard applies to this proposal first.

Walk to the nearest benchmark: 9 segments, 6 of them failing

WALK TO THE NEAREST BENCHMARK • worst per segment; red line 540 m (P5 at 0.6 m/s, 15 min)



Worst 1,111 m; for P5 that is 30.9 minutes - this proposal's own rule, and the layout as drawn does not meet it. Closing it needs 9 more third-order points, whose annual operating cost is already in this package's cost table rather than left to a next version.

Closure evidence and measured corpus

CLOSURE EVIDENCE AND MEASURED CORPUS • n=927 (git tree) • 2026-08-20

Closure evidence: the mechanism has been run

`node visual/assets/run_s08_tabletop.js` • 10 cases • 2 accepted • 8 refused • re-runnable by the reviewer • rule and reason quoted verbatim from the shipped file

- **accept** — f <= F is level for this cycle and the scenario continues
- **accept** — f > F returns the whole route, not the failing station
- **refuse** — rule 5: amending only the worst station is not an available action
  record.stations: needs at least 3 items, has 2; review parties: 2 distinct across the circuit; a reading taken by one kind of party cannot detect what that kind of party cannot see
- **refuse** — a record does not get to state its own f
  verdict.f declares 0.13 but the stations recompute to 0.360000; verdict.passed says true but f 0.360 against F 0.2 derives false; verdict.action is "continue" but f against F requires "return-whole-route"
- **refuse** — a circuit read by one kind of party cannot detect what that party cannot see
  review parties: 1 distinct across the circuit; a reading taken by one kind of party cannot detect what that kind of party cannot see
- **refuse** — without an equivalent non-AI path the stop rule would be circumvented
  record.verdict.non_ai_path_available: must be true, got false; verdict.non_ai_path_available is false: a scenario with no non-AI equivalent cannot be returned, because returning it would interrupt public service — so the stop rule would be circumvented
- **refuse** — omitting the non-AI path field is not the same as satisfying it
  record.verdict: missing required property "non_ai_path_available"; verdict.non_ai_path_available is absent: the equivalent non-AI path is not waivable, so omitting the field cannot be treated as satisfying it. State it explicitly as true.
- **refuse** — a connecting route must terminate at a first-order benchmark, or nothing checks the carried value
  stations[last]: a connecting route must terminate at a first-order benchmark, because the carried value has nothing to be checked against otherwise. This one ends at BM-302 (third).
- **refuse** — rule 7: two consecutive passes are required, because once may be luck
  resumption needs two consecutive passing cycles, record shows 1; once may be luck
- **refuse** — rule 7: an F1 incident resumes only on unanimous confirmation, no majority rule
  resumption after an F1 incident requires unanimous confirmation by all four review categories

Nothing here is a field measurement. Not one reading was taken: these are worked values chosen to exercise the refusal branches.

Measured corpus: few take this road

927 packages, read from the git tree (872 listed in the index). Pulse or lifeline is at 193 and proof or verification at 208, both saturated; measurement or levelling is at 13. Motif detection is a keyword regex and therefore a lower bound, and the convergence on three-areas-two-wings is induced by the taskbook rather than agreed.

Leveling is not another railway-operations metaphor. It is a method for how trust is established.

What is left of the belt with the AI switched off

Without a non-AI equivalent the stop rule is circumvented. All 12 cards name their equivalent; the 4 F1 ones keep theirs permanently (full text in scenario_cards.json).

S01	F3	Paper sign-in and on-site stewards	S02	F2	Residents report breaks on paper at the nearest third-order benchmark,...	S03	F2	The staffed counter runs independently of the agent
S04	F1	A staffed help desk and a printed directory of institutions, departments and clinic hours	S05	F2	Authorisation credentials and change records are documents	S06	F1	Devices stop and return to their bays
S07	F3	The evidence hall's physical display and paper archive stay open	S08	F3	Physical interpretation plates along the route and scheduled human-led tours	S09	F2	Prices and licences go on a physical notice board
S10	F1	A review meeting and a paper record	S11	F1	The test ground stops serving the public and its boundary is held by physical...	S12	F2	Checking reverts to user reports on paper or by phone plus quarterly...

Sources: this package's metrics.json and visual/assets/field_map.json, both shipped with it; the authoritative source is the GitHub git tree. The independent verifier visual/assets/verify.js checks every class-1 metric. Generation scripts are in the accompanying issue. The corpus grows daily; re-run before citing.

IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

FIG.07

a type drawing; it carries no orientation

THE LEVELING LINE

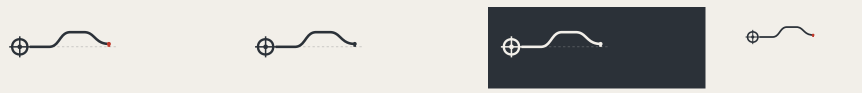
The mark draws the method, not decoration: the datum line departs the origin, rises, returns, and misses the datum - that red gap is the closure error.

Construction

- **Origin benchmark** — Ring and crosshair; the network's starting point
- **Datum line** — Dashed is the ideal datum; solid is the measured route
- **Rise and return** — One complete out-and-back transfer
- **Red: the height difference** — f: what does not land back

Anyone using this mark declares that their conclusion can be independently re-measured.
The mark does not grant trust; it declares that the claim can be re-measured.

VARIANTS



Two-colour • primary · Single-colour • etched and cast · Reversed • dark ground · Minimal • below 16px

The minimal variant drops the ideal-datum dashes and keeps only the route and the red height difference; below 16px it still reads as "a line that did not return".

APPLICATIONS

Benchmark plaque

BM-0
Origin benchmark • AI Origin Community
Annual origin and closure computation
Re-survey records publicly searchable

Reading board • L2 closure stele

closure error f, this cycle
0.26 Tolerance F3 = 0.20
Over F — S08 AI cultural guiding • RT-N
This device is allowed to look bad: publicly displaying failure builds more trust than any success narrative.

Data interface icon

- F1 Tolerance class: the mark's colour
- F2 Tolerance class: the mark's colour
- F3 Tolerance class: the mark's colour

Extension rule

Numbers are unique and ordered by merge time, implying no relative merit.

Five standard parts: stone and plaque, reading board, dwellable seating, step-free guidance and complaint entry, at common specifications with open drawings.

No unlicensed typefaces, images, trademarks or portraits are used.

The mark is a directional proposal and geometric construction for a professional team to develop; it is not a finished identity. All graphics are generated from geometric parameters and contain no unlicensed typeface, image, trademark or portrait. The 0.26 shown on the reading board is illustrative, not a measured result.

INNOVATION ECOSYSTEM AND ELEMENT MECHANISMS

FIG.08

a type drawing; it carries no orientation

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element		Spatial carrier		Benchmark		Re-survey item
Land and space	→	Key-area renewal parcels	→	BM-1 / BM-0 / BM-2	→	Delivery rate of promised space
Compute	→	Consolidated compute nodes	→	BM-1	→	Share of public compute open to application
Data	→	Data-factor circulation venue	→	BM-2	→	Authorisation-chain completeness
Scenarios	→	Real-use venues in the two wings	→	BM-21 / BM-22	→	Enforceability of scenario exit
Funding	→	Zhongguancun technology-services wing	→	BM-21	→	Deliberately blank — see the note below
Talent	→	Talent district and third places	→	BM-0	→	Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.
Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.
Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Numbering	Case	Trust mechanism	Re-measurable	Transferable point
C1	**Algorithm registers**	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	**Risk-tiered AI legislation**	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	**Standardised testing frameworks**	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	**Algorithmic impact assessment practice**	Mandatory ex-ante and ex-post review for high-risk uses	Med-high	Supports the strictest tolerance for health scenarios
C5	**Civic data-stewardship practice**	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	**Open-source reproducibility norms**	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.
A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied, no marks used, no non-public data, and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land_use.geojson and public_space.geojson.

LANDMARKS, KIT OF PARTS, SIGNAGE SYNTAX AND OPERATING CYCLE

FIG.09

a type drawing; it carries no orientation

Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

BM-0 • AI Origin Community

Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan

closure error f, this cycle
0.26 Tolerance F3 = 0.20 **Over F**
S08 AI cultural guiding • RT-N • f = 1 − consistency • illustrative

A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. **The value of this landmark is that it is allowed to look bad.**

L3 return-to-datum point

BM-2 • Dazhongsi

An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

Kit of parts for public space: five standard components

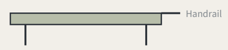
KIT OF PARTS • common specifications and open drawings, so any new node joins in one language

BM-301
Third-order benchmark plaque
Monthly re-survey • unique number

1 Stone and plaque
The number is position, order and cycle

Closure error f
0.14 Level, within tolerance
Next re-survey: March

2 Reading board
Visible on site; no internet needed



3 Dwellable seating
With armrests: older people rely on them to stand

Clear width and gradient enter the re-survey items

4 Step-free guidance
Read by the users themselves

Scan to appeal
or phone / on-site registration
A number on receipt • reply within 5 working days
No number = not accepted; counted

5 Complaint entry
A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	**Origin benchmark**	Annual origin and closure computation
BM-1 / BM-2	**First-order benchmark**	Annual re-survey
BM-2x	**2nd-order point**	Quarterly re-survey
BM-3xx	**3rd-order point**	Monthly re-survey
RT-N / RT-S	**Closing route**	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	**Community day**	third order BM-301/BM-302/BM-303	P4 • P5 • P7
Quarterly	**Scenario open day**	second order BM-21/BM-22	P2 • P3
Semi-annual	**Route re-survey**	first order, routes RT-N / RT-S	P1 • P2
Annual	**Zeroing ceremony**	L3 return-to-datum point	all four review parties present

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and $f \leqslant F$ is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.

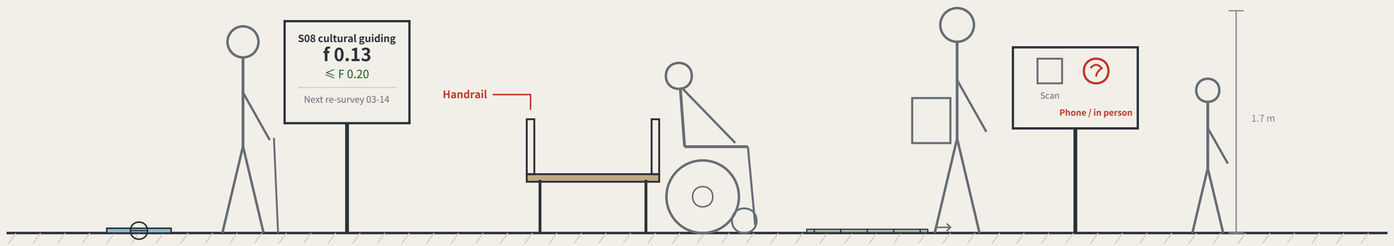
THE BENCHMARK ON THE STREET AND HOW THE KERB IS SHARED

FIG.10

a type drawing; it carries no orientation

What a benchmark looks like at eye level

A BENCHMARK AT EYE LEVEL • third-order BM-301 • elevation 1 m = 138 units



P4 older resident • P5 wheelchair user • P7 frontline worker • P6 child

1 Stone and plaque
Flush with the ground, under a metre across
Uniquely numbered • walk-up

2 Reading board
Visible on site; no internet needed
Exceedance marked in datum red with the date it was returned

3 Dwellable seating
Armrests are mandatory: older people rely on them to stand
A point is only re-measurable if people stay near it

4 Step-free guidance
Continuous clear width and gradient enter the re-survey items
Wheelchair passing is read by the user in person

5 Complaint entry
A non-scan method is mandatory
A number on receipt • reply within 5 working days

Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

The same kerb, after snow clearance

In winter the order becomes one real trade-off

Piled on the device band
Devices stop; S06 falls back to staffed transfer

Piled on the pedestrian band
Pedestrians pushed into the carriageway - the worst of the three

Piled on the kerbside parking band — This proposal takes this one
The one item in the order that may be lost first

Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER • sequence and precedence only, no widths

1. **Emergency and fire access** — Never occupied
2. **Step-free boarding, turning** — Not occupied
3. **Walking and dwelling** — Never squeezed below the level-of-service reserve
4. **Public transport and cycle parking**
5. **Device charging and standby** — Sited only from what remains after the four above
6. **Kerbside car parking**

satisfied first → … → allocated last

The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. **The plate's f 0.13 is an illustration, not a reading**: none is measured yet. Component specifications are in geometry/public_space.geojson and the scenario cards; positions await official boundaries and title verification.

REGIONAL COORDINATION INTERFACES: EXTENDING THE LEVELLING NETWORK

FIG.11

a type drawing; it carries no orientation

An executable form of coordination: how two levelling networks join

This belt's network
Departs BM-0 • F set at BM-1

The adjacent network
Own datum • own F

A joint observation produces a number, not an intention:
Each network reads the same public question set;
the difference is the cross-network closure error. Within a jointly agreed tolerance,
the two networks' records may be mutually recognised; over it, each re-surveys,
and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

Partner	Stage	Exchanges	Explicitly does not exchange	Precondition for the interface
BDA (Yizhuang)	Volume production and high-level AV road testing	Mutual recognition of closure records across complementary speed domains	Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds	Both parties confirm the speed-domain split and test protocols
Future Science City	Corporate research institutes and engineering	Scenario demand and real user density	No claim on IP; no scheduling on anyone's behalf	Wing benchmarks built and a first baseline taken
Huairou Science City	Large facilities and basic research	The measurement method itself: how f is defined, cycle length, the statistics behind revising F	No scenario data is exchanged; the coordination is methodological, not applied	The closure convention published and open to methodological review
Beiwei Community	The innovation community the taskbook names	Reciprocal joint points; exchanged review parties	This proposal does not know its positioning and does not set its benchmark order	Each party names at least one publicly reachable joint point
Beijing-Tianjin-Hebei	Cross-provincial coordination	Cross-provincial recognition of the tolerance convention	No direct recognition of records; without a shared convention they are not comparable	Tolerance definition and revision procedure agreed across provinces

None of the five rows has been confirmed by the party named.
No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.

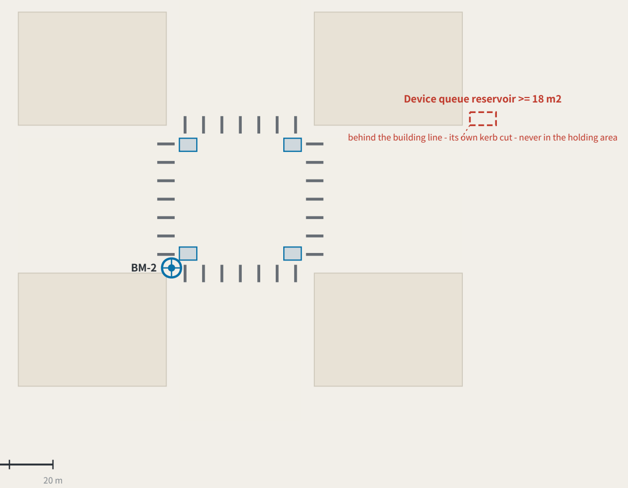
Four-quadrant pedestrian connection and device queue reservoir at Dazhongsi

FIG.12

a type drawing; it carries no orientation

I. Four-quadrant plan

FOUR-QUADRANT PLAN • TYPE APPLIED AT BM-2, NOT A SURVEY



Blue blocks are pedestrian holding areas, one per quadrant; the red dashed outline is the device queue reservoir, set behind the building line with its own kerb cut.

If the four quadrants are not connected, devices and pedestrians must contend for the same holding area — which is why this district names it the most concrete spatial task it has.

2. Dimensions: design intent / as-measured

DIMENSIONS: DESIGN INTENT vs AS-MEASURED

Item	Design intent	As measured
Pedestrian holding area, per quadrant width >= 3.0 m x depth >= 4.0 m	⩾ 12 m²	
Crossing effective clear width after deducting poles, covers and the furnishing strip	⩾ 4.0 m	
Kerb ramp gradient Barrier-Free Environment Construction Law and current codes	⩽ 1:12	
device queue reservoir set behind the building line, with its own kerb cut	⩾ 18 m²	
Reservoir to holding area, clear distance must not cross any desire line	⩾ 2.0 m	
BM-2 stone to pedestrian edge the stone is flush with the ground and is not a trip hazard	⩾ 0.6 m	

The right-hand column is deliberately empty.
Capacity is computed on site from measured clear width. A number filled inwithout measuring is manufactured certainty. Completed before any device is admitted.

3. Kerb allocation section (order and intended widths only)

KERB ALLOCATION SECTION

(1) Frontage	(2) Through walking	(3) Furnishing strip (BM-2 sits here)	(4) Kerb
2.5 m	**3.0 m**	**1.8 m**	**0.7 m**

Building line to kerb: frontage, walking, furnishing, kerb.BM-2 sits in the furnishing strip, so a reading is never taken in the walking zone.

This is a type drawing applied at BM-2, not a survey of the existing junction. Every dimension is design intent and the as-measured column is deliberately empty: carrying capacity must be computed on site from measured effective clear width. Junction geometry, channelisation and signals rest with the road authority and a traffic study.

The three key areas in section, at one scale

FIG.13

One horizontal scale • 220 m across each • no vertical exaggeration • ±0.00 is that area's own benchmark, not a shared absolute datum

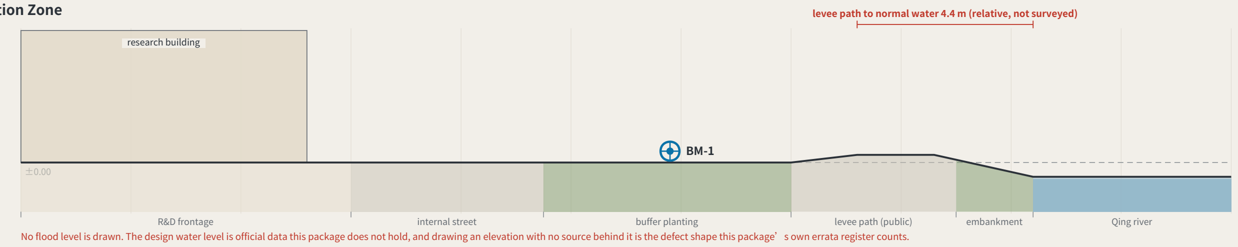
a type drawing; it carries no orientation

Zhongzhiyuan AI Acceleration Zone

±0.00 = BM-1

192.9 ha • BM-1
dominant use R&D 100%
2 land-use classes inside

External arrival. The area is 100% single-use, so public access to the Qing river runs through the block or not at all — what this section sets out is a public route across the R&D frontage and onto the levee path.



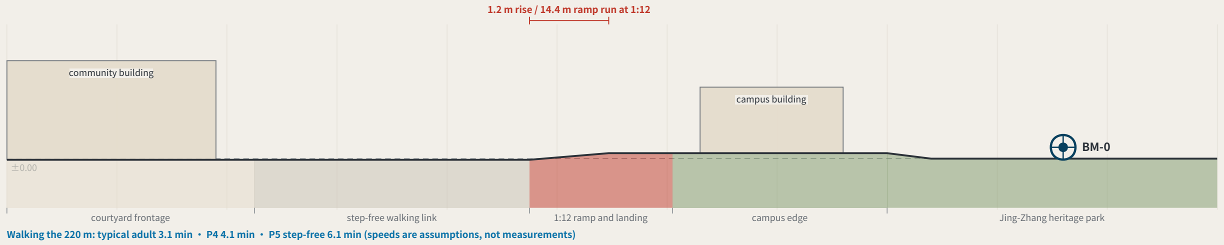
No flood level is drawn. The design water level is official data this package does not hold, and drawing an elevation with no source behind it is the defect shape this package's own errata register counts.

Beijing AI Origin Community

±0.00 = BM-0

104.3 ha • BM-0
dominant use community services 94%
2 land-use classes inside

A 1.2 m grade change, and step-free continuity. The campus edge ends in steps, and P5 needs a continuous step-free route. The reason this section exists is to put a 1:12 ramp with a landing where the steps are.



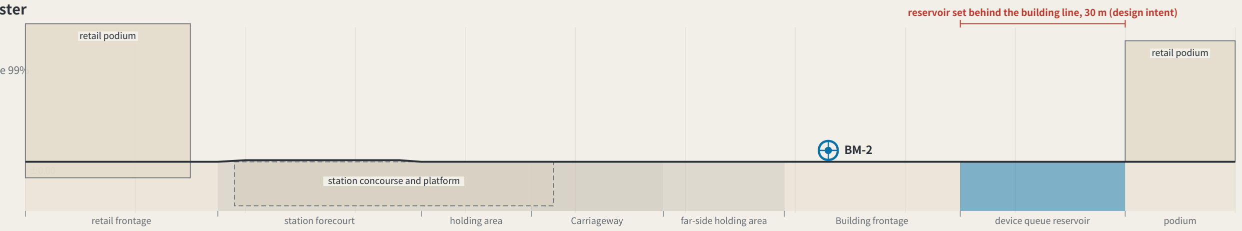
Walking the 220 m: typical adult 3.1 min • P4 4.1 min • P5 step-free 6.1 min (speeds are assumptions, not measurements)

Dazhongsi AI Industry Cluster

±0.00 = BM-2

72.0 ha • BM-2
dominant use industry and commerce 99%
2 land-use classes inside

A station below grade and a queue behind the building line. The only section here with anything below ground, and the only one with a carriageway. The device queue reservoir sits behind the building line with its own kerb cut, clear of the crossing desire lines — the four-quadrant plan is FIG.12.

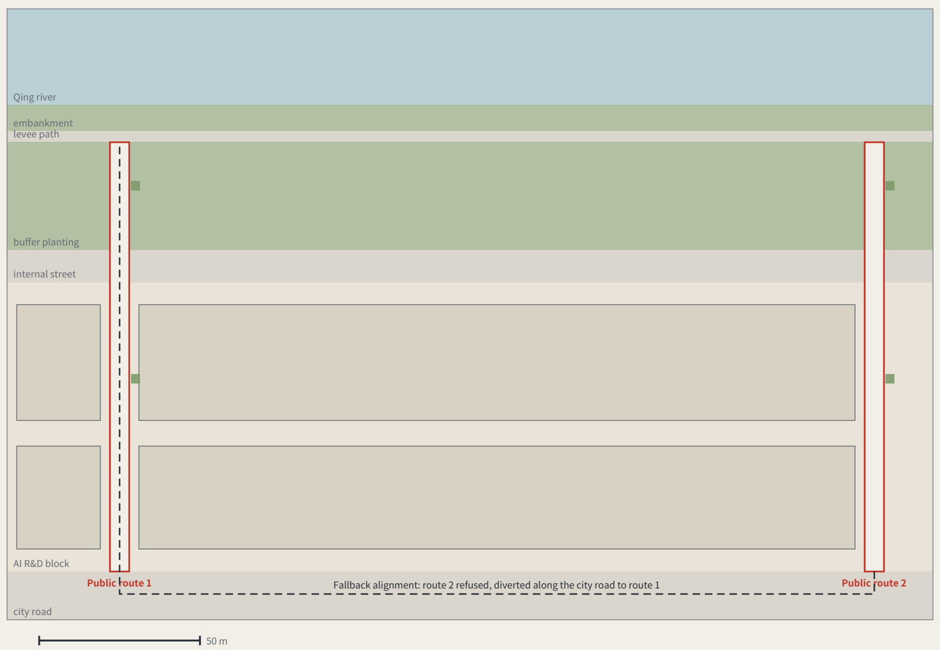


One horizontal scale, one datum convention, no vertical exaggeration. They are placed together so the charge that the three are one template used three times can be tested: one reaches a river across a 100% single-use block, one is step-free continuity over a 1.2 m rise, one is a station below grade with the device queue behind the building line. ±0.00 is each area's own benchmark — absolute elevations need official levelling data this package does not hold, so they are not drawn. Concept design intent, not surveyed profiles.

The through-block public route to the Qing river

FIG.14

a type drawing; it carries no orientation



The route along its length: city road to levee path

2.6× the scale of the plan • no vertical exaggeration • 133.5 m rising 1.4 m, an average of 1:95

The steepest stretch is inside the planting, and 1:20 is the constraint on that stretch — an average gradient would hide it.

Dimensions: design intent and as built

The right column is deliberately empty — effective clear width is filled in on site

Item	Design intent	As measured
Public route clear width two abreast, and a wheelchair passing	⩾ 6.0 m	
Route centre spacing a 939.0 m frontage needs 4 routes, 234.8 m apart	⩽ 250 m	
Longitudinal gradient 133.5 m rising 1.4 m, an average of 1:95, step-free throughout	no stretch steeper than 1:20	
Opening hours a route that can be locked is not public access; it is permission	24 h, ungated	
Spacing of places to sit standard component KIT-03, with armrests	⩽ 60 m	
Levee path clear width two-way walking, passing	⩾ 3.5 m	
Building setback from the route the ground floor may not face it with a solid wall	⩾ 3.0 m	

The count is not a preference. It is a division.

Under the provisional substitute boundary the northern frontage of Zhongzhiyuan runs 939.0 m. At a maximum centre spacing of 250 m that is 4 public routes, 234.8 m apart. Replace the boundary with an official polygon and the number moves with it — it is computed from the geometry at build time, not written onto the sheet.

What one gate costs

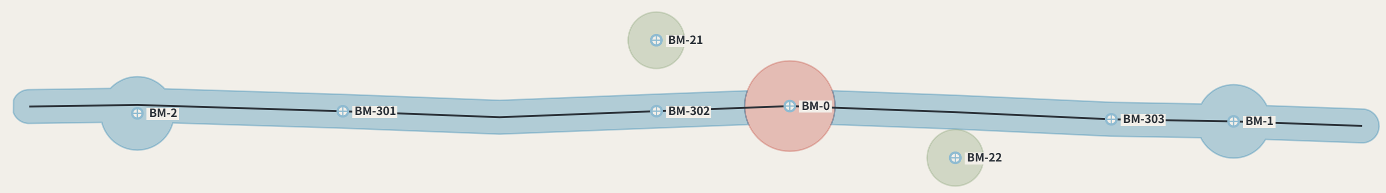
With the route open it is 133.5 m from the city road to the levee path. With route 2 refused and the walk diverted along the city road to route 1, it is 368.3 m. For P5, walking at 0.6 m/s, 3.7 minutes becomes 10.2 minutes. That is the working unit of a public route that can be locked.

FIG.13's section says this edge needs a public route across the R&D block onto the levee path, and no drawing showed it. This is that route: 6.0 m clear, centres at most 250 m apart (4 of them across a 939.0 m frontage), open 24 hours and ungated. The fallback alignment and the cost of a refusal are drawn. A type drawing, not a survey; dimensions are design intent, the as-measured column is empty; boundaries are provisional.

Phasing: advanced by closure results, not by dates

FIG.15

north is right (unrolled)



Phases advance on closure results, not on years

Phase	Increment	Spine added	Benchmarks this phase adds	Condition that opens it (not a date)	Re-surveys/yr	Annual running cost
Near	32.1 ha	0.64 km	BM-0	none — no prior closure needed	1	**CNY 2,484–7,922**
Mid	234.8 ha	8.80 km	BM-1, BM-2, BM-301, BM-302, BM-303	four near-term items done; f ⩽ F ×2	39	**CNY 13,864–45,492**
Long	25.1 ha	—	BM-21, BM-22	mid phase: f ⩽ F ×2	47	**CNY 19,444–63,922**

The table above phases the eight points that exist and their cumulative cost. FIG.21 shows that meeting the fifteen-minute rule needs 9 more third-order points (a further CNY 22,005–76,358 a year), falling by order into the mid and long phases — this is not the phasing of a network that already complies.

The first phase is a stone, a plate and one reading a year

The near-term increment is 32.1 ha and holds BM-0 alone: 1 re-survey a year, CNY 2,484–7,922 to run. That line is the answer to "this mechanism is too heavy" — it can start from one point, and that point costs what a sub-district office can approve by itself.

The condition is a number, not a year

The mid phase does not begin because a second year arrived. It begins because the four near-term items are complete and f ⩽ F for two consecutive cycles; the long phase the same. Once may be luck — the same rule this proposal writes at PATH-4 of the conversion path, applied to phasing. Fail it and the programme stays where it is, which makes the phasing table an exit table as well.

Which four are the four near-term items

The condition that opens the mid phase reads "the four near-term items complete" and nothing in the package says which four — and the proposal contains two different sets of four: the near-term projects R1–R4, and the ones needing no official data, R1–R3 and R8. The second set contains a long-term item, so reading the gate against it makes the gate circular. Taking the first:

R1 datum stone L1 and the public evidence hall — one complete closure and its reading published within the first cycle
R2 first public setting of the tolerance F — a first version of F1/F2/F3 each, posted
R3 the four-week closure trial on cultural guidance S08 — closure recomputed and the consistency ratio published within four weeks
R4 third-order benchmarks set out at communities and rail stations — one recorded reading at each of the three community points

Two naming errors this sheet found

Dropping each benchmark into the phase polygons contradicts two of the phase names. The mid phase was named for "three first-order points" and contains two (BM-1, BM-2); the third is the datum BM-0, which is in the near phase. The long phase was named for "the full three-order network", and the three third-order community points already fall inside the mid phase — what the long phase actually adds is the two second-order wing access points. The names were written for the intent and never checked against the polygons. Both are corrected in the layer, and every cell of this table is computed from the geometry rather than typed.

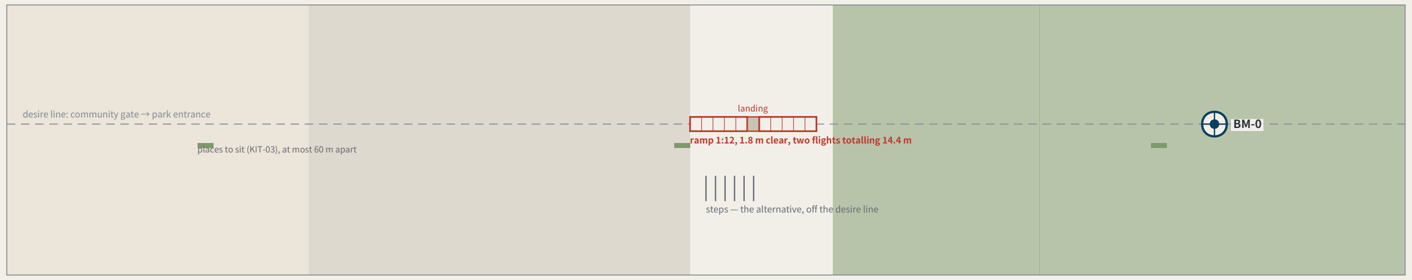
phasing.geojson has shipped for a long time, been repaired once and enters the recompute path, and no drawing had ever shown it. This sheet draws the three increments and costs each phase's year with the same model as build_operations, not a second set of coefficients. Phases open on closure results, not on years. The geometry is concept advice on provisional boundaries and must be recomputed when official ones are published.

The step-free link from the origin community to the heritage park

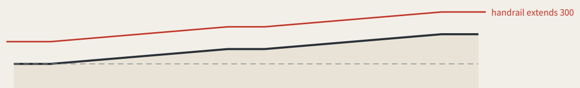
FIG.17

a type drawing; it carries no orientation

I. Plan



II. The ramp along its length (no vertical exaggeration)



rising 1.2 m in two flights of 7.2 m with a 1.5 m landing between

5 m

What offsetting the ramp would cost

The usual arrangement puts the steps on the desire line and offsets the ramp. Offset it by 22 m and whoever uses it walks 44 m further every time — out and back. For P5, at 0.6 m/s, that is 1.2 minutes a trip; at twice a week for a year, 4.6 km. Nobody on the steps walks a metre further for the ramp being on the line: steps are shorter, so they sit beside it.

Walking the 176 m

typical adult	**2.4 min**
P4	**3.3 min**
P5	**4.9 min**

speeds are assumptions, not measurements

Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

Item	Design intent	Why it is that number	As measured
Ramp gradient	**1:12**	step-free throughout; the ramp is on the desire line and the steps are its alternative	
Ramp clear width	**⩾ 1.8 m**	two people passing, or a wheelchair with someone alongside	
landing	**one every 750 mm of rise, at least 1.5 m long**	a 1.2 m rise split into two flights — there has to be somewhere to stop	
Handrail	**both sides, extending at least 300 mm past each end**	a handrail that stops at the top of the slope stops where it is still needed	
Level approach at each end	**at least 1.5 m level**	level ground before and after, so nobody has to turn on the slope	
Spacing of places to sit	**⩽ 60 m**	standard component KIT-03, 450 mm high with armrests	
Steps	**beside the ramp, off the desire line**	anyone who prefers steps is not sent around — steps are shorter than the ramp	

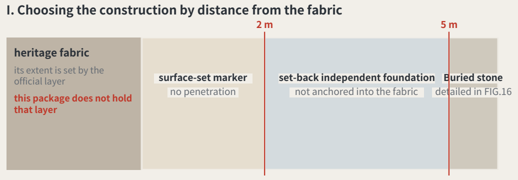
FIG.13' s section says this place exists to put a 1:12 ramp where the steps are, and no drawing in the package had shown it. One design decision: the ramp sits on the desire line and the steps beside it — offset the ramp by 22 m and its user walks 44 m further every trip, while nobody on the steps walks a metre more. A type drawing, not a survey; dimensions are design intent and the as-built column is empty.

Monuments beside heritage fabric: the no-drilling construction and its setback rule

FIG.19

a type drawing; it carries no orientation

I. Choosing the construction by distance from the fabric



d = horizontal distance to the outer edge of the fabric

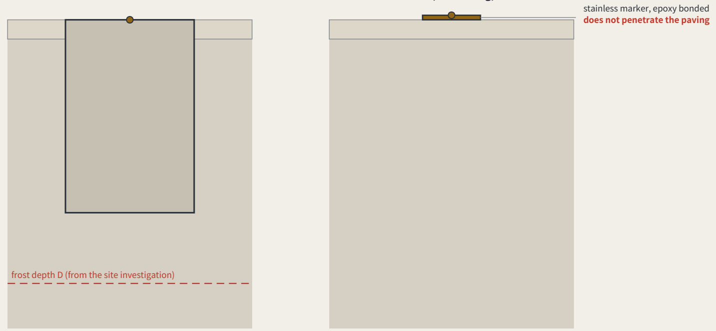
2 m

The rule is complete; its input is missing

Which line d is measured from depends on the officially published heritage protection area and its construction-control zone. This package does not hold that layer: each of the 3 shipped constraint features records it as a data gap in its own note, and this proposal does not infer it. So the sheet gives the rule, the thresholds and the three constructions, and leaves that line to whoever publishes it — guessing a protection boundary to make the drawing look complete would mean inventing the one thing here that is not this proposal' s to decide.

II. Two constructions, one scale

Buried: foundation below the frost line **Surface-set: bonded, no drilling, no excavation**



can carry any order; removal needs plant

moves with the paving through the year → third order only, tied monthly

500 mm

III. What each construction costs

Construction	Applies when	Foundation	Orders it may carry	Re-survey cycle	What removal leaves
Buried stone	d ⩾ 5 m	below the standard frost depth D + 100 mm	datum / first / second / third	per order	plant to lift out; leaves an excavation
set-back independent foundation	2 ⩽ d < 5 m	below frost line, set back ⩾ 2 m, unanchored	second / third	per order	plant to lift out; leaves an excavation
Surface-set marker (no penetration)	d < 2 m	no foundation; bonded to existing paving, no drilling	**third order only**	**monthly, and tied to the nearest buried point each time**	lifting it leaves only a bond mark, which cleans off

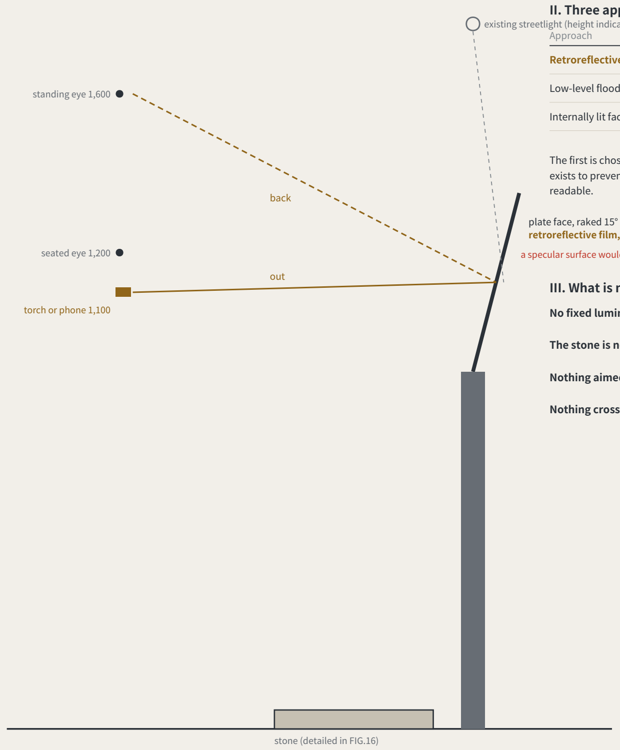
The call is for a belt along a hundred-year-old railway; this proposal' s central act is sinking concrete. Three constructions, chosen by distance from the fabric, with the cost stated: without excavation there is no foundation below the frost line, so the surface-set marker may not carry first- or second-order duty. Where d is measured from depends on the official heritage layer, absent here and recorded as a gap in constraints.geojson.

Reading after dark, without lighting the benchmark

FIG.20

a type drawing; it carries no orientation

I. Geometry: where the light comes from, and returns to



II. Three approaches, compared by how they fail

Approach	Needs power	How it fails	Light trespass
Retroreflective face, reader brings the light	no	the face gets dirty → visible in daylight → caught before it stops being readable	none
Low-level floodlight	yes, at every point	the lamp dies → the plate looks intact and cannot be read	yes, needs controlling
Internally lit face	yes, at every point	the lamp dies → the whole face goes, and it still looks like a plate	yes, and aimed at the reader

The first is chosen not because it is cheap but because its failure is visible. When a lamp dies the plate still looks intact and can no longer be read — exactly the thing this proposal exists to prevent, a measurement whose failure cannot be seen. A dirty retroreflective face is simply dirty, visible in daylight to anyone walking past, well before it stops being readable.

III. What is not lit

No fixed luminaire	the reader brings the light; no power supply, no luminaire to maintain
The stone is not lit	it sits flush with the ground; lighting it adds no legibility, only night-time glare
Nothing aimed upward	no supplementary light may be emitted above the horizontal
Nothing crossing the boundary	the added vertical illuminance at neighbouring dwelling windows must meet the local limit — that figure comes from the local standard, which this package does not hold

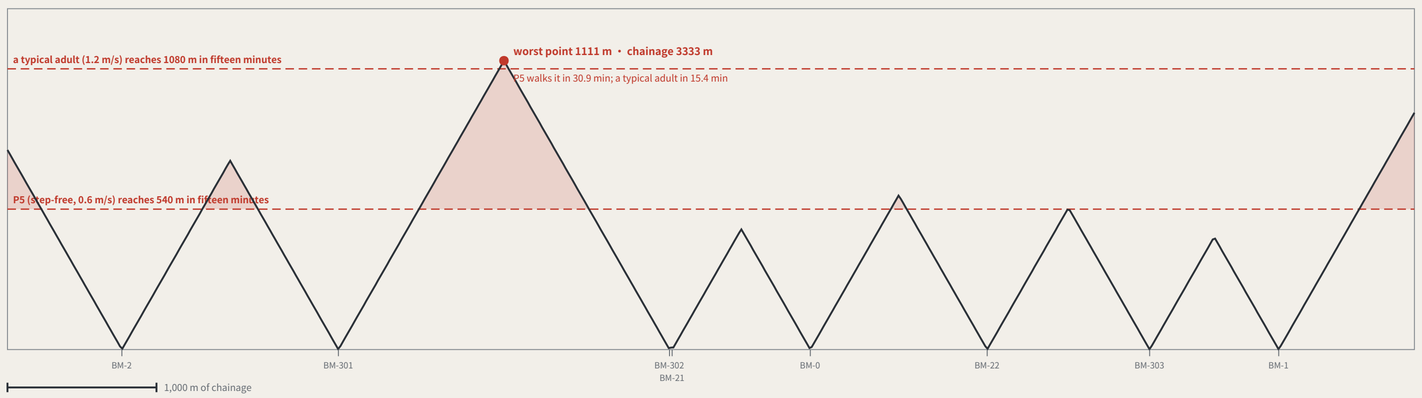
Three third-order points are read monthly by residents, and much of the year those days fall after dark — and the package had never said how the plate is read at night. The obvious answer is a lamp, and it is the wrong one: it needs power, needs maintaining, throws light into the windows opposite, and when it dies the plate looks intact and cannot be read — a measurement whose failure is invisible, which is the thing this proposal exists to prevent. So the face is retroreflective with no fixed luminaire and the reader brings the light; a dirty face is simply dirty, in daylight.

How far the nearest benchmark actually is: this proposal's own rule, applied to itself

FIG.21

a type drawing; it carries no orientation

I. Distance to the nearest benchmark, along the spine



II. Turning that sentence into a spacing rule that can be checked

Where the rule comes from	this proposal's own repeated claim: a right of review you must walk fifteen minutes to reach has not been given
Interior spacing	chainage between adjacent points ⩽ 2 × 540 m = 1080 m
Both ends of the line	end of line to its nearest point ⩽ 540 m
Which speed the rule uses	P5's 0.6 m/s, not a typical adult's 1.2 m/s — setting the rule by the faster figure sets it at a standard most people cannot use

III. How far short the current layout falls

6 of the 9 segments between the 8 existing points fail. Closing the gap needs 9 more third-order points (or 3 if the rule is only held to an average walker). They are not free: third-order points are read monthly, and every one of them enters the annual operating table. **This sheet does not place them** — putting 9 coordinates on a provisional substitute spine is the manufactured certainty this register counts. The rule ships; the positions wait for the official alignment.

This proposal argues repeatedly that a right of review you must walk fifteen minutes to reach has not been given, and its entire participation argument rests on anyone being able to walk up to a benchmark and take a reading — and nobody had ever measured that walk. Measured: the worst place on the line is 1111 m from the nearest benchmark, 30.9 minutes for P5, twice the limit this proposal holds others to. The rule ships as a spacing requirement; the positions wait for the official alignment.

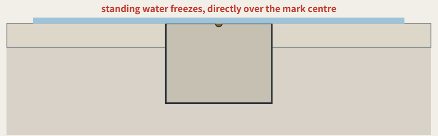
Winter: water, ice, and two of this package's rules in collision

FIG.22

a type drawing; it carries no orientation

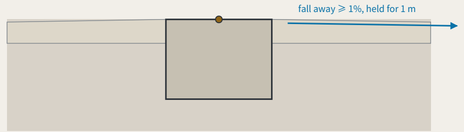
I. Two ways of doing it, at one scale

Wrong: level paving, no fall



Level paving with no fall: compliant in September, the only ice in the sidewalk in January

Right: the stone as a local high point



The stone is the local high point: flush with the paving beside it, and that paving falls away

1,000 mm

The fall is drawn true: 1% over 1 m is 10 mm, which is barely visible at this scale — and a slope nobody can see on a drawing is a slope nobody checks on site, while every dimension on the sheet is still satisfied.

II. What that requires of the construction

Item	Requirement	Why
Stone top against the paving	**flush, tolerance ±5 mm**	KIT-01: no trip hazard
Fall around it	**⩾ 1%, away from the stone, held for ⩾ 1 m**	makes the stone a local high point — water leaves instead of gathering
Protective ring	**none**	a recessed ring is a bowl; a raised one is the trip hazard KIT-01 forbids
Surfacing	**permeable or tight-jointed; no joint channel running inward**	a joint channel leads water to the mark centre
Winter reading	**taken after clearing, with the condition recorded**	a cleared reading and an uncleared one are not directly comparable

III. Three things a winter reading adds

Clear it — clear snow and ice from the stone and 300 mm around it before reading
a reading taken over a film of ice is measuring the ice

Check the drainage — confirm the stone is still the high point: no ponding marks or ice within 1 m
a ponding mark is evidence the drainage has already failed, and it appears before the ice

Record the condition — mark the cycle record as a winter condition, and whether the stone was cleared
cleared and uncleared readings are not directly comparable — unrecorded, nobody knows whether they should be

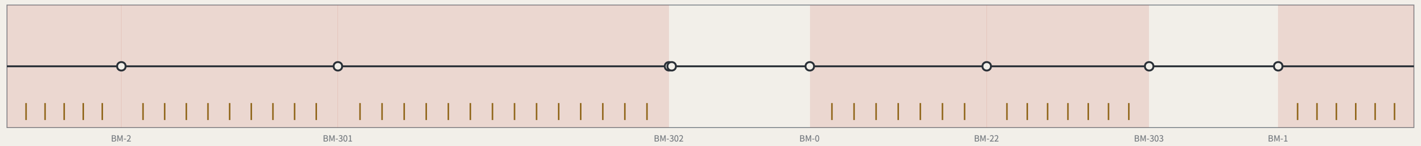
The three third-order points are read monthly, January included, and this package had never said what that costs. KIT-01 requires the stone flush with no trip hazard and FIG.16 gave that ±5 mm — but a mark flush in level paving sits in standing water: the tolerance that makes it safe in September makes it dangerous in January. The answer is a local high point, not a ring.

Finding the nearest benchmark: which way, and how far

FIG.23

north is right (unrolled)

1. Wayfinding only in the segments where this proposal is already in breach



Red marks the segments FIG.21 judged to fail the fifteen-minute rule; one wayfinding post every 150 m, 48 in all. Compliant segments get none — putting up a sign where it is already close enough is decoration.

1,000 m

2. The four things the plate must answer

← BM-302
410 m • P5 approx. 11 min

BM-0 →
520 m • P5 approx. 14 min

Standard part KIT-04, no new structures

Direction — **← BM-302 / BM-0 →**
Either end of the line may be nearer; only at the midpoint are both sides equally far

Distance — **410 m / 520 m**
metres, because metres can be checked by the person who walks them

Time — **P5 approx. 11 min / 14 min**
At 0.6 m/s, the same figure as in every other drawing in this package (personas.json)

Numbering — **BM-302 • BM-0**
A reading taken on arrival is attributed by it; no need to ask afterwards where the person was standing

3. What wayfinding cannot do

These 48 wayfinding posts will not make 1111 m any shorter. What they do is let a person know before setting out how far it is, which way, and what the point is called on arrival. This is mitigation, not a fix. The fix is the 9 third-order points in FIG.21, which give no position until the official alignment is published. Writing the wayfinding up as the solution is exactly the kind of substitution this package has kept catching in itself — using one small thing it can do to cover one large thing it cannot.

FIG.21 measures the farthest point on the whole line at 1111 m from the nearest benchmark: 9 segments, of which 6 fail this proposal's own fifteen-minute rule. Closing the gap takes 9 third-order points, but their positions await the official alignment. People still have to find the points that already exist. This drawing supplies wayfinding: placed only inside the segments already in breach, one every 150 m, 48 in total, using the existing standard part KIT-04, adding no new structures. Wayfinding will not make 1111 m any closer — this is mitigation, not repair.

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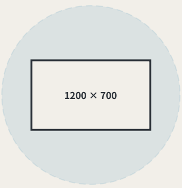
The device envelope: what exactly goes in those 18 m²

FIG.24

a type drawing; it carries no orientation

1. Envelope and turning, plan

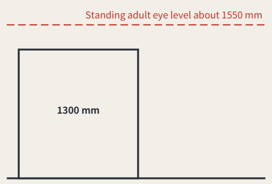


Turning diameter 1800 mm

1,000 mm

Step-free clear walking width. Devices passing, parked or queuing may not squeeze it below this figure — the space for people is set aside first, and devices get only what is left (FIG.12).

2. Elevation: why the height is this number



Height sits below eye level so queued devices do not cut the sightline across the sidewalk. **No external basis; set by this proposal** — eye height varies from person to person, and the sheet gives the reasoning, not a clause to copy.

3. The envelope items, and the constraint source of each

Item	Value	Constraint source	Why it is that number
Length × Width	**1200 × 700 mm**	**published by this proposal**	Back-calculated from the standalone kerb cut width in FIG.12: a device that cannot pass the opening is not accepted by this proposal
Height (with load)	**1300 mm**	**Set by this proposal**	Taken below the sightline of a standing adult, so that queuing devices do not block eye contact across the sidewalk; no external basis, set by this proposal
Turning diameter	**1800 mm**	**Set by this proposal**	Required to turn in place; it sets the shape of the reservoir space and not just its area — precisely the half that "18 m²" cannot state
Kerb weight	**⩽ 120 kg**	**Set by this proposal**	Taken at the order a single person can push clear after a power failure; beyond that, mechanical assistance is required, and there is nowhere to place it at a kerb cut
Site speed	**⩽ 6 km/h**	**Set by this proposal**	Speed ceiling in the shared pedestrian-vehicle zone, taken at the order of walking speed; no citable local regulation was obtained, so it is registered as a self-set value
Clear width left after passing	**⩾ 1500 mm**	**published by this proposal**	Step-free clear walking width, the same value as KIT-04; equipment passage may not press it below this number
Marking	**Cap height ⩾ 25 mm**	**published by this proposal**	ID number, responsible body and appeal channel, all three present; same rule as the reading plate: attribution starts with a visible ID number (FIG.16, FIG.20)

4. Converting 18 m² into a number of units — the two drawings must agree with each other

One standing bay is the envelope length times the turning circle: 2.16 m². FIG.12 publishes a 18 m² floor for the reservoir, so that is **8 at once**. The number did not exist before: "⩾ 18 m²" never said how many, and two devices of equal footprint with different turning radii do not need the same reservoir. The build checks the two sheets against each other — if the area does not divide into bays, or the count exceeds what the floor holds, it refuses.

FIG.12 fixed the device reservoir at ⩾ 18 m² and this package had never drawn a device. Area is not specification: equal footprints with different turning radii need different reservoirs. An envelope, not a product; 7 dimensions each marked with what bounds them, 4 bounded by nothing but this proposal — **attack those first**. It holds 18 m² / 8 devices.

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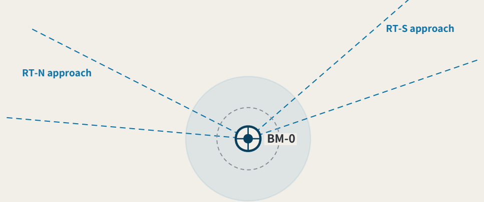
BM-0, the origin: the place both routes have to return to

FIG.25

north is up

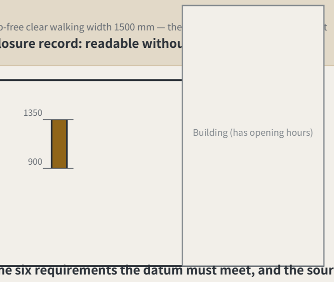
1. Datum site: instrument setup circle, clear walking width, approach sightlines of the two routes



Instrument setup circle ⌀ 2400 mm

Step-free clear walking width 1500 mm — the

2. Closure record: readable withou



A reading is published the moment it is taken, yet the evidence is public only as far as the door. So the closure record sits **outside the door**, facing the sidewalk, on the plate geometry FIG.16 fixes (eye height 900–1350 mm). Inside the hall it would turn anyone can walk up and read into during opening hours.

3. The six requirements the datum must meet, and the source of each

Item	Requirement	Why, and where it comes from
Instrument setup circle	**Diameter 2400 mm**	Tripod spread 1,200 mm, operator outer circle 600 mm; closing a route means setting the instrument up on this stone, and this circle is the ground it actually takes
and the step-free clear width	**⩾ 1500 mm clear outside the setup circle**	Instrument setup may not eat the clear width — FIG.24' s rule for equipment, no slack for our own people
Route intervisibility	**One approach sight cone each; no trees or parking inside**	RT-N and RT-S terminate here, which needs sight of the previous turning point; without it you add one, and each adds a height difference
Closure record orientation	**Faces the sidewalk, eye height 900–1350 mm, read without entering**	Published as taken, but evidence is open only as far as the door — a datum with opening hours is shut when the reading lands
Display content	**Network closure error**	Network closure error only; a single-point failure shows there (erratum E50). Moving bad news back to the datum is the recorded defect
Fall	**⩾ 1% out**	As FIG.22: the stone must be a local high point, or in January it is the only ice here

BM-0 appears on nearly every sheet here and had never been drawn — and what it carries is a spatial property: **a reading is published the moment it is taken, and evidence is only as public as the door.** Hang the record in a hall with opening hours and the origin is shut at the moment the reading lands. This sheet gives the setup circle, an approach sightline cone for each closing route, and the closure record outside the door facing the sidewalk.

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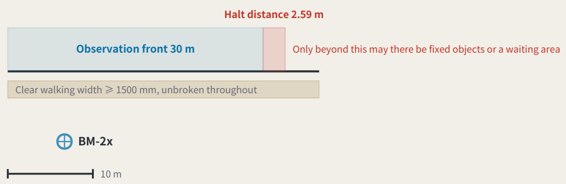
BM-2x, the second order: a scenario has to be stoppable, and that takes ground

FIG.26

north is right (unrolled)

1. Observation front and halt distance, same scale



2. This distance is calculated, not decreed

FIG.24 fixes the site speed at ⩽ 6 km/h = 1.667 m/s. Halt distance = reaction leg + deceleration leg = $1.667 \times 1 + 1.667^2 \div (2 \times 1.5) = 1.67 + 0.93 =$ **2.59 m**. **The reaction time and the deceleration are both set by this proposal, not code values** — logged in the register format as `A-DEVICE-002`, with the derivation, the consequences of a wrong value, the closing condition and the ownership, submitted together with this drawing. The formula is published with the drawing: change one set of values and this stretch of ground can be recomputed at once.

3. Where second-order and third-order points differ

Item	Value	Why
stone	**Same KIT-01 as the third-order point**	Second order and third order differ in re-survey interval and route role, not in construction — a more important place gets no special benchmark
Re-survey cycle	**Quarterly**	The two wings are the source of scenario density; readings are taken quarterly, priced on the same table as the annual first order and the monthly third order
Observation front	**30 m**	The stretch of kerb the scenario is actually watched from. "The scenario takes its reading here", given without a length, is the same kind of sentence as "reservoir space 18 m^2" given without devices
Halt distance	**2.59 m**	From FIG.24: a site speed of ⩽ 6 km/h gives reaction 1 s over 1.67 m, then deceleration at 1.5 m/s^2 over 0.93 m
Clearance requirement	**No fixed objects and no waiting area within ⩾ 2.59 m ahead of the front**	A scenario that cannot stop is not a scenario, it is an accident; this stretch of ground is the physical form of the words "it can stop"
Clear walking width	**⩾ 1500 mm, uninterrupted throughout**	The same number as in FIG.24 and FIG.25 — observation, instrument setup, equipment passage: none of them may eat into it

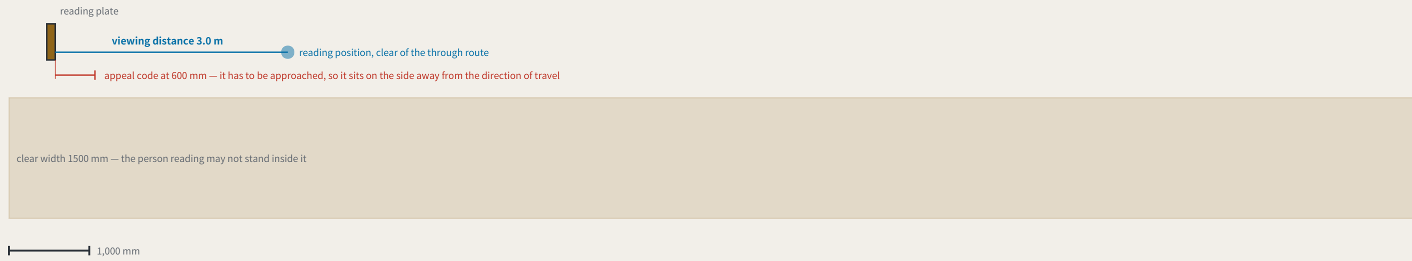
The two wings carry the second-order points; they appear in the naming table, the cost table and the resource table, and have never been drawn. What separates second order from third order is not the stone — both are KIT-01 — but that what runs here is a **scenario aimed at real users**, so it must be able to stop. Computed from FIG.24 at ⩽ 6 km/h, the halt distance is 2.59 m; within that stretch of ground ahead of the front there may be no fixed objects and no waiting area. The reaction time and the deceleration are both set by this proposal and are registered as A-DEVICE-002.

One reading: where a person stands, and whether they block the way

FIG.27

a type drawing; it carries no orientation

I. Where the person reading stands



II. This viewing distance comes from a ratio this package already published

The body text long ago checked its own drawings against "legible cap height ≈ viewing distance ÷ 250", and found the minimum cap height across all drawings, 4.73 mm, compliant. **That same ratio has never been pointed at the plate face.** Pointed at it, what it returns is not a cap height but a standing position: cap height 12 mm, viewing distance 3.0 m, and 3.0 m lies outside the 1500 mm clear walking width — the person reading and the person walking past neither yield to nor obstruct each other. Take a cap height of 6 mm and the viewing distance becomes 1.5 m, putting the reader squarely in the middle of the path: **the plate face is still legible, the sidewalk no longer works**. The 12 mm cap height is set by this proposal and is registered as `A-PLATE-001`. **This package carries a second cap height**: FIG.24 requires device markings ⩾ 25 mm, legible at 6.25 m by the same ratio — not a contradiction, and not an accident: device markings exist to identify the machine **coming toward you**, and must be read at a distance that still leaves time to react; the reading plate is read standing in front of it. Different distances, different cap heights — and with no stated relation between the two, a reader will simply assume one of them was set arbitrarily. The ratio 250 is not self-set; it is the one this package already applies to itself.

III. The four steps of one reading, and the ground each takes

Step	What happens	Why that dimension
Walk there	FIG.21 measured the worst walk at 1,111 m — 30.9 minutes for P5 — and this proposal does not meet its own rule	The cost of this step has been measured, and it measures as non-compliant
Stand	Reading at 3.0 m from the plate face, standing outside the clear walking width	Cap height 12 mm × viewing distance ratio 250 = 3.0 m; take the cap height as 6 mm and the viewing distance drops to 1.5 m, putting the person reading in the middle of the path
Read	Read the number on the plate and the benchmark_id	The number makes this reading attributable; no need to ask afterwards where the person was standing
Disagree	Scanning the appeal code requires approaching to about 600 mm	This step has to come within close range, so the stacking goes on the side facing away from the direction of travel — the one step that must intrude intrudes on the edge of the path, not its centre

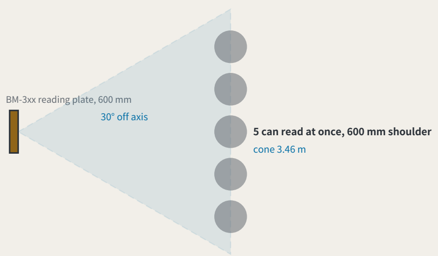
The whole participation argument rests on a single sentence — anyone can walk up to a benchmark and take a reading — yet no drawing has ever drawn the four minutes that person spends standing there. Standing is not free: the ground he occupies is the ground others walk on. Working backwards from the "legible cap height ≈ viewing distance ÷ 250" already published in this package, a cap height of 12 mm gives a viewing distance of 3.0 m, which falls just outside the 1500 mm clear walking width; take 6 mm and the reader stands in the middle of the path. The cap height is set by this proposal and registered as A-PLATE-001.

Third order: the most numerous tier, and more than one reader

FIG.28

north is up

I. The legible cone, and how many can read at once



II. The part ground cannot solve

A class of 30 cannot read at once. 5 fit the cone, so 6 rounds, and at 40 s a reading that is about 4 minutes. **The number is here because widening the ground does not solve it.** No cone holds thirty people; the rest take turns, and turns cost time — a cost that, unwritten, gets discovered on site in the form of "this point is awkward". The off-axis angle and the shoulder width are chosen by this proposal and registered as `A-READ-001`; the plate width, the viewing distance and the clear width come from FIG.16, FIG.27 and FIG.24 and are not fixed here.

clear width 1500 mm — a queue runs parallel to the kerb, never pushing into this

1,000 mm

III. Standing rules for a third-order point

Item	Value	Why
Re-survey cycle	**Monthly**	third order is the most numerous tier and the one residents actually stand at
Legible cone	**3.46 m**	a 600 mm plate (FIG.16) stays legible within 30° off axis, and at the distance FIG.27 fixes, 3.0 m, the cone opens to this width
Readers at once	**5**	cone ÷ a 600 mm shoulder. **This is this sheet answering the question FIG.27 left open** — that sheet fixed one person's position
A class of thirty	**6 rounds, about 4 minutes**	past that count the extra people are not handled by ground but by taking turns — **and turns are a time cost, which belongs in writing rather than being discovered on site**
Which way a queue runs	**parallel to the kerb, never perpendicular to the direction of travel**	a perpendicular queue pushes its tail into the 1500 mm clear width; a parallel one does not
What is not provided	**no railing, no dedicated waiting area**	a third-order point sits in a neighbourhood, and fencing it stops it being something you can read in passing; what this sheet gives is a standing rule, not a new structure

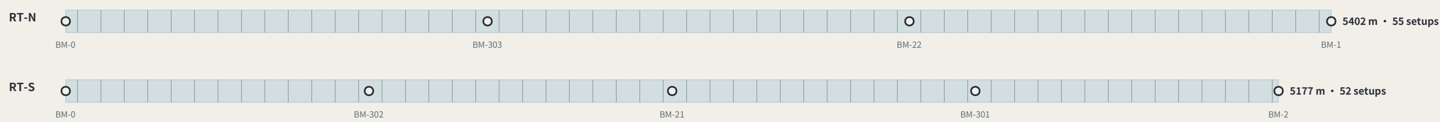
The origin and the second order each have a sheet; **the third order is the tier nobody drew, and the one residents actually stand at**. It also inherits what FIG.27 left open: that sheet's 3.0 m distance is **one person's** position. From a 600 mm plate and a 30° off-axis limit the cone is 3.46 m wide and 5 people read at once; a class of 30 takes 6 rounds, about 4 minutes — **that part is solved by turns, not by ground**.

The annual first-order closure: how long a route actually takes

FIG.29

north is right (unrolled)

I. The two closing routes, and how many setups each takes



Each vertical line is a setup, spaced by the 50 m sight — the instrument stands midway between two staves, so one setup advances the run by twice the sight. **A closure is not one act but this many**, and each is a place it can go wrong, which is exactly what the closure error measures.

1,000 m

II. The duration derived from geometry, against the band the cost table publishes

Route	Length	Instrument setup	Observing	Walk back	Total
RT-N	5402 m	55	4.6 h	1.3 h	**5.8 h**
RT-S	5177 m	52	4.3 h	1.2 h	**5.5 h**

the band the cost table publishes for a first-order session: 4.0–6.0 h

III. These two numbers had never met

The 4.0–6.0 h for a first-order point in `operations.json` was written into the model by hand, on its author's judgement, and nothing could check it. The route lengths are not hand-written: they come out of the shipped geometry. Divide the length by the viewing distance for the instrument setups, multiply by the minutes each takes, add the walk back to the datum along the spine, and the result is 5.5–5.8 h — **inside that band, and tight against its upper edge**. The build asserts it: lengthen the geometry or edit the band and the two stop being compatible and the build stops. The viewing distance and the minutes per setup are chosen by this proposal and registered as `A-SURVEY-001`; the route lengths, the walking speed and the hour band are all already published here.

The first order was the last tier without a sheet, and it is not really a place but **an event**: once a year a closing route is run, and this proposal's measure of trust comes out of that run. From the shipped route lengths, a 50 m sight and 5 minutes a setup, the two routes need 52–55 setups and 5.5–5.8 h — **inside the 4.0–6.0 h the cost table sets by hand** — and the build asserts the two stay compatible.

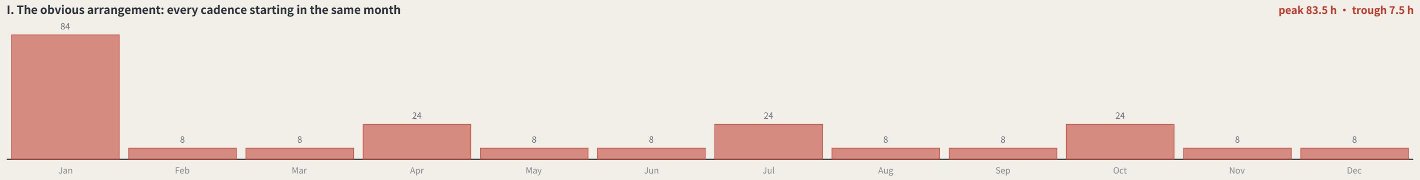
The year: where forty-seven readings fall

FIG.30

a type drawing; it carries no orientation

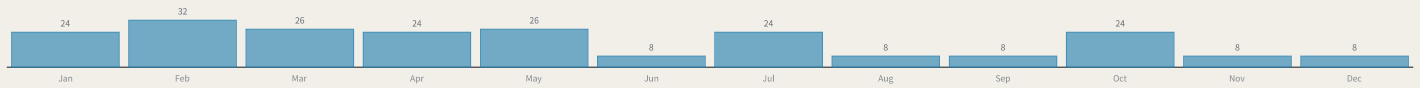
I. The obvious arrangement: every cadence starting in the same month

peak 83.5 h • trough 7.5 h



II. Separated: the same sessions, the same hours, the same table

peak 31.5 h • trough 7.5 h



III. This is a calendar problem, not a resourcing one

Three third-order points monthly, two second-order quarterly, three annual points once each — 8 points, 47 readings. Start every cadence in the same month and the year is not flat: **83.5 h falls in one month against 7.5 h in each of eight others**, eleven to one. The annual table says 139–220 h, which is exactly right and completely unable to say this. A team sized for the average cannot do that month; a team sized for it is idle for eleven. Separating the three annual readings into three months drops the peak to **31.5 h** — and the search corrected the guess that produced this sheet: intuition said stagger the quarterly points, and the search leaves them where they are and separates the annual ones instead. The reason is the per-session hours: a datum session is 24 h and a third-order one 2.5 h, so the peak is made almost entirely by annual points landing together, and moving quarterly points around a peak that size changes nothing. **The work does not change at all**: the same 47 sessions, the same hours, the same cost table. The arrangement is searched rather than asserted — every offset combination evaluated, the lowest peak taken — and if separating them did not materially beat the obvious arrangement the build would refuse, because a sheet proposing a schedule that does not help is a sheet arguing for itself.

FIG.29 priced **one** closure and the cost table prices **a year** — and no sheet had drawn the year. Drawn, it shows that with every cadence starting in the same month the peak is 83.5 h against a 7.5 h trough, eleven to one: a team sized for the average cannot do the peak month, and one sized for the peak is idle the rest of the year. Separating the three annual readings drops it to 31.5 h with **no change to the work at all**. This is a calendar problem, not a resourcing one.

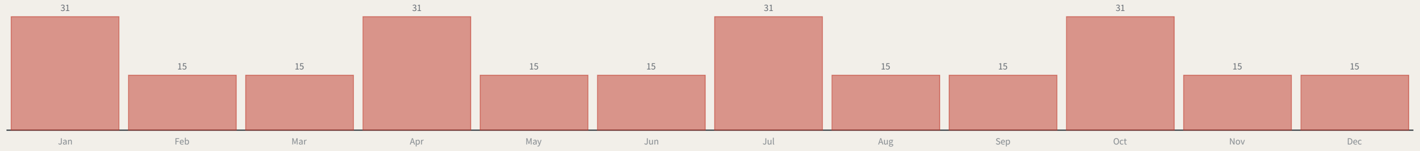
The other year: the hours this network asks of people it does not pay

FIG.31

a type drawing; it carries no orientation

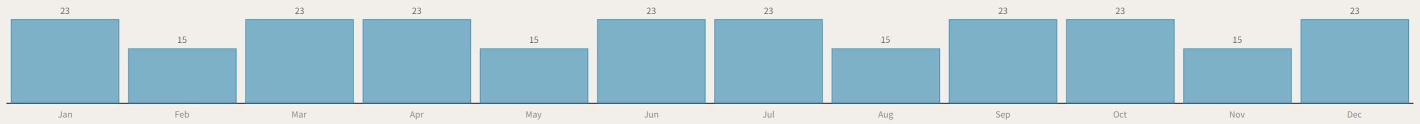
I. Under FIG.30 arrangement: the paid year is flat, the volunteer year is not

peak 31.0 h • trough 15.0 h



II. Separating the quarterly points: the volunteer peak comes down

peak 23.0 h • trough 15.0 h



III. This is not free, and the price is stated here

Volunteers do not attend every session: the datum and first-order runs are professional work with no community places, and community readings fall on the **second and third order** — exactly the tier FIG.30 had no reason to move. So the arrangement that flattens paid hours leaves the volunteer year at 31.0 h against a 15.0 h trough, while that sheet stated that moving the quarterly points changes nothing — **true of the quantity it was measuring and false of this one**. Separating them brings the volunteer peak to 23.0 h at the cost of raising the paid peak from 31.5 h to 33.5 h. **No arrangement minimises both.** Drawing two profiles without saying which is taken hands the decision to the reader while looking rigorous; **this proposal takes the volunteer peak**. A paid peak is a procurement problem — hire, subcontract, move a date. A volunteer peak is a participation failure: community readers cannot be scaled to a month, and that they turn up at all is the floor the whole argument stands on. The 2.0 h added to the paid peak is the price of that choice, stated rather than absorbed.

FIG.30 flattened **paid** hours and said the quarterly points need not move. Volunteers attend only the second and third order — precisely the tier left alone — so the volunteer year stayed at 31.0 h against 15.0 h. Separating them brings it to 23.0 h at a cost of 2.0 h on the paid peak. Both cannot be minimised at once, and this proposal takes the volunteer peak: a paid peak is a procurement problem, a volunteer peak is a participation failure.

How many people: the cheapest roster is the one that empties the instrument

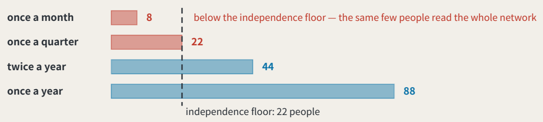
FIG.32

a type drawing; it carries no orientation

I. Eighty-eight attendances, and where they fall

Benchmark	Order	Attendances/yr	Min. distinct readers
BM-21	second	8	2
BM-22	second	8	2
BM-301	third	24	6
BM-302	third	24	6
BM-303	third	24	6
Total		88	22

II. Roster size depends on how often each person is asked



III. The cheapest arrangement is the one that empties the instrument

Community readings fall on the second and third order: **88 attendances** a year, 148–244 h. The cheapest cover is 8 people at one attendance a month — **and that is exactly the failure `A-CLOSURE-002` describes**: this package's own most fragile assumption states that if the parties are not independent the closure error comes out **systematically small**, and the mechanism reports success having measured nothing. It fails invisibly, and in the direction that looks like success. **So the cheapest staffing answer is the one that empties the instrument.** This sheet turns that into a number: **no community reader is asked for more than 4 attendances a year** — not chosen here but read from `personas.json`, where P3 is the mainstay of the quarterly open days and P8 executes the quarterly re-survey, and nobody is described as coming more often. A third-order point runs 24 attendances and so needs 6 distinct readers; a second-order point runs 8 and needs 2. The network floor is **22 distinct community readers** — not a target but a **floor**, below which the independence this mechanism rests on stops being arguable. An earlier version of this sheet chose that cap as half a point's sessions**, which put 4 readers on a third-order point at 6 visits each — every two months,** more than this package's own personas say they give. Inventing a number when a citable one already exists is the shape this register keeps recording; cited instead, `A-ROSTER-001` now registers a reference rather than a chosen ratio.

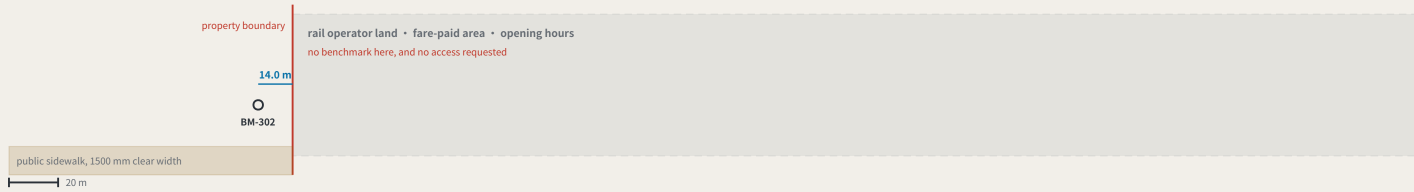
FIG.31 scheduled volunteer **hours** and never asked how many **people** that is. The year needs 88 community attendances, and the cheapest cover is 8 people once a month — **which is exactly the failure A-CLOSURE-002 sets out**: parties that are not independent make the closure error systematically small, and the mechanism reports success having measured nothing. The rule: nobody attends more than half a benchmark's readings in a year, giving a network floor of 22 people.

Dazhongsi: where the belt meets ground this network does not own

FIG.33

north is up

I. The interface: the point stands outside controlled ground



The 14.0 m is not a rule invented here: the prose already fixes a 14 m setback for buildings along the spine, and the benchmarks sit on the axis, so the nearest point stands clear of controlled ground as a consequence. The build checks that no benchmark falls inside any building footprint — one that did would be a reading that needs a ticket.

II. Handover: what crosses the boundary is the reading, not the instrument

Item	Rule	Why
What crosses	**the reading, not the instrument**	the operator levels their own point on their own ground, to the same tolerance class, and hands over a number
What does not	**access, equipment, working hours**	this proposal does not ask for access it cannot be granted, nor pretend a point it cannot reach is a point it holds
If they decline	**the segment is recorded unread**	not estimated, not interpolated, not assumed within tolerance — an unread segment is a visible hole in the closure record
What the hole means	**that segment may not be used to claim compliance**	**which is precisely why this package built a closure record that can show a hole**
Where a point may stand	**never on controlled ground**	if a reading needs a ticket, "anyone can walk up to a benchmark" is false as stated

Every other sheet assumes this network reaches its own points; at the station that ends — the landowner is not the city. Three rules: **no benchmark on controlled ground** (BM-302 stands 14.0 m clear of the hub, checked at build time); **what crosses is the reading, not the instrument**; and **if the operator declines the segment is recorded unread** — not estimated, not interpolated, a visible hole in the closure record.